

Stabilizing Actor-Critic Learning for Stochastic Disaster Relief Logistics under Partial Observability

Abstract

Effective disaster response requires sequential resource allocation under partial observability, evolving infrastructure conditions, and stochastic disruptions, conditions in which static optimization methods often struggle to adapt. This study investigates actor-critic reinforcement learning for large-scale humanitarian logistics under uncertainty and operational constraints. It introduces a multi-district disaster-response simulation environment, compatible with standard reinforcement-learning interfaces, that models stochastic demand evolution, terrain-aware accessibility changes, multimodal transportation constraints, inventory dynamics, and disruption-driven hazards across disaster scenarios in Nepal and Thailand. The proposed artificial intelligence method is a post-decision actor-critic approach built on proximal policy optimization. It combines autoregressive action factorization, feasibility-aware masking, recurrent belief aggregation, curriculum-controlled stochasticity, and dual-critic post-decision value estimation to improve stability in high-dimensional sequential decision processes. The application of artificial intelligence is disaster relief logistics, where the method supports the allocation of humanitarian resources under changing operational conditions. Experiments compare the proposed method with greedy and rolling linear-programming baselines under stochastic environment dynamics and deterministic policy decoding. They also include multi-seed, reward-sensitivity, and ablation analyses. Results show higher delivered-help performance, improved reward stability, lower delay penalties, and favorable operational-efficiency trade-offs at comparable deployment costs. Feasibility-aware masking and belief-guided accessibility estimation provide the most consistent practical benefits. Overall, the work provides a reproducible simulation framework and evidence that structured actor-critic reinforcement learning can support stable and effective humanitarian logistics decisions under uncertainty.

Keywords:

Artificial Intelligence, Reinforcement Learning, Actor-Critic Learning, Proximal Policy Optimization, Disaster Relief Logistics, Humanitarian Decision Support,

1. Introduction

Disasters, whether triggered by natural hazards or human activity, pose complex and high-stakes challenges for humanitarian logistics. In the immediate aftermath, aid agencies must respond to sharp, spatially heterogeneous spikes in demand while contending with damaged infrastructure, disrupted supply chains, and uncertain transport conditions. These difficulties are especially pronounced during the critical 72-hour window, where delays in delivering food, water, medical aid, or shelter can lead to irreversible suffering [1]. Traditional approaches based on static optimization, rule-based dispatch, or centrally planned heuristics typically assume fixed network conditions and stationary demand [2, 3]. As a result, they struggle to adapt when severity levels evolve, routes become blocked, or priorities shift across regions.

Moreover, the underlying transition dynamics are highly uncertain due to aftershocks, evolving infrastructure damage, uncertain accessibility, and fluctuating operational demand. Field evidence shows that effective response often emerges from adaptive, decentralized coordination rather than rigid top-down command [4]. Recent work has also explored deep reinforcement learning for post-disaster infrastructure recovery and resilience-oriented operational planning under uncertainty [5]. Technically, early reinforcement-learning applications such

as Q-learning for post-disaster routing or resource allocation [6, 7, 8] demonstrate potential but struggle with unstable learning, delayed feedback, and poor scalability in multi-zone, dynamically evolving environments. These issues are further exacerbated by abrupt changes in accessibility, evolving demand, and the high-variance nature of policy-gradient estimation in stochastic environments [9].

Reinforcement learning (RL) provides a principled framework for sequential decision-making under uncertainty [10] particularly through policy-gradient and actor-critic methods, which combine value-function baselines with stochastic policy updates [11]. Proximal Policy Optimization (PPO) offers additional stability by constraining policy updates through a clipped surrogate objective [12]. However, applying PPO-style actor-critic algorithms to disaster logistics poses unresolved challenges: (i) designing environments that capture realistic multi-district spatial and logistical dynamics (ii) mitigating high variance induced by disruption-driven accessibility and evolving demand and (iii) ensuring that policies remain feasible while balancing efficiency, fairness, and urgency.

To address these limitations, we develop a unified experimental framework for evaluating sequential decision-making under uncertain multi-district disaster conditions. The framework models evolving accessibility, disaster severity, opera-

tional constraints, and resource demand over time, enabling reinforcement-learning agents to be evaluated in settings that more closely reflect real post-disaster operations than simplified routing formulations. Finally, we propose a structured actor-critic framework referred to as Post-Decision Proximal Policy Optimization (PD-PPO).

The auxiliary Post-Decision critic is conceptually motivated by Post-Decision state formulations in Approximate Dynamic Programming (ADP), where value estimation is performed after deterministic action effects but prior to stochastic exogenous transitions, thereby simplifying stochastic value approximation under uncertainty [13, 14].

We formulate disaster response as a partially observable decision-making problem with stochastic transition dynamics governing system evolution across districts and time. Within this setting, we investigate whether modern policy-gradient methods can learn resource-allocation strategies that remain reliable under uncertainty, adapt to shifting operational priorities, and remain applicable across geographically distinct scenarios under re-training.

Experimental results indicate that optimization-based and heuristic baselines remain competitive under relatively stable operating conditions, but exhibit greater sensitivity to stochastic disruptions and operational variability. In contrast, the proposed framework exhibits stable behavior and better trade-offs across multiple objectives under uncertainty.

Our main contributions are as follows:

- We introduce MultiDistrictDisasterEnv, an OpenAI Gym-compatible [15] based on a Partially Observable Markov Decision Process (POMDP) [16, 17] simulation environment that captures multi-district disaster dynamics, including stochastic accessibility, evolving demand, and multi-modal transportation constraints across realistic geographic settings.
- We provide a formal specification of the environment dynamics, including stochastic transition models for severity, demand, and transport accessibility, enabling reproducible and controlled experimentation under partial observability.
- We propose a structured actor-critic reinforcement learning framework that combines autoregressive action factorization [18, 19, 20], feasibility-aware action masking, belief-informed accessibility estimation, and dual-critic Post-Decision value estimation to improve training robustness and sequential decision consistency under uncertain disaster conditions.
- We design a multi-objective evaluation protocol assessing efficiency, fairness, and timeliness, and benchmark reinforcement learning policies against Greedy and Rolling Linear Programming (Rolling LP) baselines under stochastic disaster conditions using deterministic policy evaluation.
- We provide a systematic empirical analysis of structured reinforcement learning components in large-scale, par-

tially observable, and combinatorial logistics environments through multi-seed ablation and sensitivity studies.

The remainder of this paper is organized as follows. Section 2 reviews related work on humanitarian logistics and reinforcement learning. Section 3 presents the proposed disaster-response environment and PD-PPO framework. Section 4 reports experimental results, ablation studies, and sensitivity analysis. Finally, Section 5 concludes the paper and discusses future research directions.

2. Related Work

Disaster logistics research spans decades of work in operations research, humanitarian supply chains, and increasingly data-driven approaches. Classical studies grounded in network flow theory provided early insights into the structure of distribution and routing decisions under constrained infrastructure [2]. Subsequent empirical analysis of large-scale crises such as the 2015 Nepal earthquake highlighted how dynamic changes in population needs, damaged transportation networks, and evolving hazard conditions complicate the timely delivery of relief [21, 22]. These real-world challenges have motivated a shift from static planning toward adaptive, uncertainty aware decision systems capable of responding to non-stationary demand, fragmented accessibility, and heterogeneous vulnerability across affected regions. Within this broader landscape, research has expanded from deterministic optimization to metaheuristics, stochastic modeling, and more recently, reinforcement learning approaches for improving responsiveness and operational effectiveness in humanitarian settings.

Network Optimization and Classical OR Models: Traditional research in disaster logistics has been dominated by network optimization and operations-research formulations. Early work modeled relief distribution, routing, and facility location as network flow or mixed-integer programming problems (R. Ahuja et al. [2]; R. Tomasini and L. Van Wassenhove [3]). These approaches offer strong structural guarantees but assume deterministic demand, complete network observability, and static planning horizons conditions rarely satisfied in real disaster scenarios where accessibility and demand evolve unpredictably.

Metaheuristics for Multi-Objective Disaster Allocation: To address the computational intractability of exact optimization, later studies adopted metaheuristics such as genetic algorithms, particle swarm optimization, and nature-inspired heuristics for multi-objective allocation and routing (L. Yu et al. [23]; A. Majumder et al. [7]). Although scalable and flexible, these methods remain dependent on hand-crafted decision logic and cannot adapt continuously from interaction with dynamic or partially observable environments.

Reinforcement Learning for Humanitarian and Emergency Logistics: A growing line of work has introduced reinforcement learning into disaster and humanitarian contexts. (C. Cheng et al. [6]) applied Q-learning to post-disaster transportation planning under suffering-driven objectives, while (A. Majumder et al. [7]) proposed a multi-

objective RL framework balancing efficiency, equity, and urgency. RL-based allocation has also been explored in epidemic response (B. Chen et al. [24]), humanitarian distribution networks (L. Yu et al. [23]), microgrid restoration (H. Nie et al. [25]), and emergency responder stationing via multi-agent RL (A. Sivagnanam et al. [26]). Despite promising results, most prior work relies on value-based or single-step policy methods (e.g., Q-learning, Deep Q-Network (DQN)) that suffer from high variance, delayed feedback, and limited scalability in multi-zone stochastic environments.

Actor-Critic Methods and the PPO Family: Actor-critic algorithms combine the strengths of policy gradients with value-function baselines to achieve stable, sample-efficient learning in sequential decision-making (I. Grondman et al. [27]; V. Mnih et al. [28]). Proximal Policy Optimization (PPO) constrains policy updates through a clipped surrogate objective, improving robustness in noisy or partially observable environments (J. Schulman et al. [12]). Extensions such as dual-critic PPO and post-decision actor-critic specifically target variance reduction in stochastic settings (E. Greensmith et al. [9]), while resource-aware exploration strategies such as Resource-Aware Exploration Strategies (RAEB) dynamically adjust exploration pressure under scarcity (Z. Wang et al. [29]). However, these methods are typically validated on benchmark control tasks rather than realistic geospatial logistics domains.

Environment Modeling for Disaster RL: Environment design plays a central role in determining RL performance. Classical dynamic programming emphasizes the importance of realistic transition models and structured value functions (W. Powell [13]; D. Bertsekas [14]). In the RL domain, potential-based shaping (Ng et al. [30]), dynamic shaping (Devlin et al. [31]), and curriculum learning (Y. Bengio et al. [32]) are widely used to reduce reward sparsity and improve training stability. Disaster-focused RL studies incorporate urgency, suffering minimization, and equity using multi-objective reward formulations (A. Majumder et al. [7]; C. Cheng et al. [6]), while RAEB-like bonuses further adapt exploration based on resource scarcity (Z. Wang et al. [29]). Still, most existing environments abstract away critical factors such as multi-modal transport, stochastic accessibility, depot stock depletion, and district-level severity evolution.

Reward Design for Humanitarian Objectives: Reward functions in disaster logistics must jointly balance efficiency, fairness, urgency, and responsible resource usage. Prior work employs weighted reward functions (A. Majumder et al. [7]), epidemic or suffering-based penalties (B. Chen et al. [24]), and priority-weighted reward terms (Yang et al. [37]). Although effective in controlled scenarios, these formulations often depend on simplified environment dynamics and do not fully capture the temporal, spatial, and operational complexity of real disaster responses.

In contrast to these prior environments and reward formulations, our work couples a realistic, multi-modal disaster logistics simulator with a stability-focused PPO-based actor-critic architecture. The next section introduces MultiDistrictDisasterEnv and details how we model severity transitions, accessibility, inventory, and multi-objective rewards in a unified PO-

MDP framework.

3. Methods

3.1. Environment Design

We model humanitarian response as a sequential resource allocation problem over a disaster-affected region in Nepal. The primary testbed consists of $Z = 12$ districts in Bagmati Province, each represented as a geographic polygon derived from Geographic Information System (GIS) shapefiles and overlaid on an OpenStreetMap (OSM) road network [22]. Districts differ in population, remoteness, prosperity, health access, and hazard exposure. These attributes are grounded in World population counts and qualitative assessments of terrain and infrastructure conditions [33, 21]. To ensure reproducibility, these attributes are parameterized using normalized indices and fixed scaling constants, allowing consistent reconstruction of district-level characteristics across simulation runs. The single-depot assumption is adopted to isolate allocation and accessibility dynamics without introducing additional coordination variables associated with multi-depot routing. Episodes span 50 decision steps, intended to emulate the critical response window after a large-scale event [1].

3.1.1. Geo-spatial Topology and Transport Modes.

The road network is extracted using OSMnx Python library and represented as a directed graph with weighted edges capturing motorable travel distances between districts. Precomputed shortest-path distances from the depot to each district are updated online when intermediate districts become blocked. Air deliveries are modeled using great-circle distances between the depot and district centroids, with terrain- and weather-dependent grounding of sorties. Districts are further categorized into plains, valley, and hilly classes, which modulate both hazard transition probabilities and travel-time slowdowns (e.g., landslide-prone hill districts experience more frequent and longer closures).

3.1.2. POMDP Formulation.

Formally, the environment is modeled as a partially observable Markov decision process (POMDP)

$$\mathcal{E} = (\mathcal{S}, \mathcal{A}, \mathcal{O}, P, R, \gamma), \quad (1)$$

where $\mathcal{S}$ is the latent state space, $\mathcal{A}$ the action space, $\mathcal{O}$ the observation space, $P(x_{t+1} \mid x_t, a_t)$ the transition dynamics, $R(x_t, a_t, x_{t+1})$ the reward function, and $\gamma \in (0, 1)$ the discount factor [16].

- zone-resource demand $D_{z,r,t}$ for $z = 1, \dots, Z$ and resources $r \in \{\text{food, medicine, water}\}$;
- depot stocks $S_{r,t}$ and per-step dispatch caps;
- zone conditions $c_{z,t} \in \{\text{Normal, HeavyRain, Landslide, Both}\}$;
- severity $h_{z,t} \in \{1, 2, 3\}$ and priority $p_{z,t} \in \{1, 2, 3\}$;

- soft beliefs $b_{z,t}^{\text{road}}, b_{z,t}^{\text{air}} \in [0, 1]$ that each mode is open to the zone z ;
- cumulative lives helped $H_{z,t}^{\text{cum}}$ and time since the last successful service.

Latent State Definition.. The underlying system state at time t is defined as

$$x_t = (D_t, S_t, c_t, h_t, p_t, A_t, H_t), \quad (2)$$

where D_t denotes zone–resource demand, S_t depot stock levels, c_t zone conditions, h_t severity levels, p_t priority levels, A_t accessibility indicators, and H_t cumulative service history.

The agent does not observe the full latent state. Instead, at each decision step it receives a factored observation vector $o_t = O(x_t) \in \mathbb{R}^d$ composed of:

- normalized demand per zone and resource, using adaptive normalization based on exponentially weighted historical statistics;
- normalized depot stock and per-resource dispatch capacity;
- severity indicators rescaled to $[0, 1]$;
- global constraints, including remaining air sorties and feasibility flags for depot access;
- belief exponential moving average (EMA)s for road and air openness, updated from the observed success or failure of past trips;
- service-history EMAs, encoding how recently and how well each zone has been served.
- mode-specific blocked-rate EMAs and attempt-rate EMAs for road and air access;
- optional expected-openness features for road and air access computed from short-horizon history.

This yields an observation of dimension $Z \cdot R + 2R + 10Z + 3$ for $Z = 12$ zones and $R = 3$ resources and is designed to be scalable to other configurations.

3.1.3. Action Space and Feasibility Constraints.

At each timestep, the agent selects a set of K dispatch decisions from the factored action space

$$a_t^{(k)} = (z, r, q_{\text{bin}}, t_{\text{trip}}, m),$$

where z is the target zone index, r the resource type, q_{bin} a discrete quantity bin, t_{trip} the number of parallel trips, and $m \in \{\text{Road}, \text{Air}\}$ the transport mode. The joint action is therefore

$$a_t = \{a_t^{(1)}, a_t^{(2)}, \dots, a_t^{(K)}\}.$$

Multiple actions can be issued per step, subject to:

- per-resource dispatch caps and global fleet limits;

- an air-sortie budget (e.g., up to three air missions per step);
- depot-stock feasibility (no overdrawn of inventory).

These actions are jointly constrained by capacity and feasibility limits:

$$\sum_{k=1}^K U_r(a_t^{(k)}) \leq S_{r,t}, \quad \forall r \in \mathcal{R}, \quad (3)$$

$$\sum_{k=1}^K \mathbb{1}[m^{(k)} = \text{Air}] \leq M_{\text{air}}, \quad t^{(k)} \leq t_{\text{max}}, \quad (4)$$

where $U(\cdot)$ denotes resource usage and M_{air} is the air sortie limit.

If an attempted dispatch becomes infeasible due to dynamic accessibility or stock constraints, the environment applies a constrained feasibility-resolution step before execution. This step may substitute transport mode, redirect the dispatch to a reachable zone with compatible unmet demand, or convert the attempt into a failed dispatch with penalty.

3.1.4. Demand, Hazard, and Accessibility Dynamics.

Initial demand $D_{z,r,0}$ is synthesized from a vulnerability score that combines remoteness, prosperity, health access, and hazard exposure, together with discrete severity and small multiplicative jitter. Demand is capped by population-dependent maxima to avoid unrealistic values. Over time, demand grows or decays based on unmet needs and past services, with higher-severity, high-need zones accumulating demand more quickly:

$$D_{z,r,t+1} = \max\left(0, D_{z,r,t} + \eta_1 \phi(h_{z,t}) \cdot \text{need}_z - \eta_2 U_{z,r,t} + \epsilon_{z,r,t}\right), \quad (5)$$

where $U_{z,r,t}$ denotes units delivered, $\phi(h_{z,t})$ is a severity-dependent growth factor, and $\epsilon_{z,r,t}$ represents stochastic demand fluctuations. In implementation, demand growth is additionally bounded by population-scaled ceilings and per-step growth caps to maintain numerical stability and realistic humanitarian demand levels. The stochastic components of the simulator, including condition transitions, cooldown sampling, aftershock dynamics, air-grounding uncertainty, and demand initialization noise, are formally summarized in Appendix G.

Zone conditions $c_{z,t}$ evolve through terrain-aware stochastic transition rules with state-dependent cooldowns, distinguishing between plains, valleys, and hilly regions. Stochastic aftershock events can temporarily increase severity and induce additional landslides. Accessibility is determined by a reachability function $\text{accessible}(z, m)$: road access is blocked in landslide states, while air may be grounded under heavy rain or combined hazards. A dynamic road-distance cache updates shortest paths to avoid blocked intermediate districts when possible; otherwise, zones become unreachable by road.

For each terrain category, condition transitions are governed by a terrain-specific stochastic transition matrix

$$P_c^{(\tau)}(i, j) = \Pr(c_{z,t+1} = j \mid c_{z,t} = i, \tau_z = \tau), \quad (6)$$

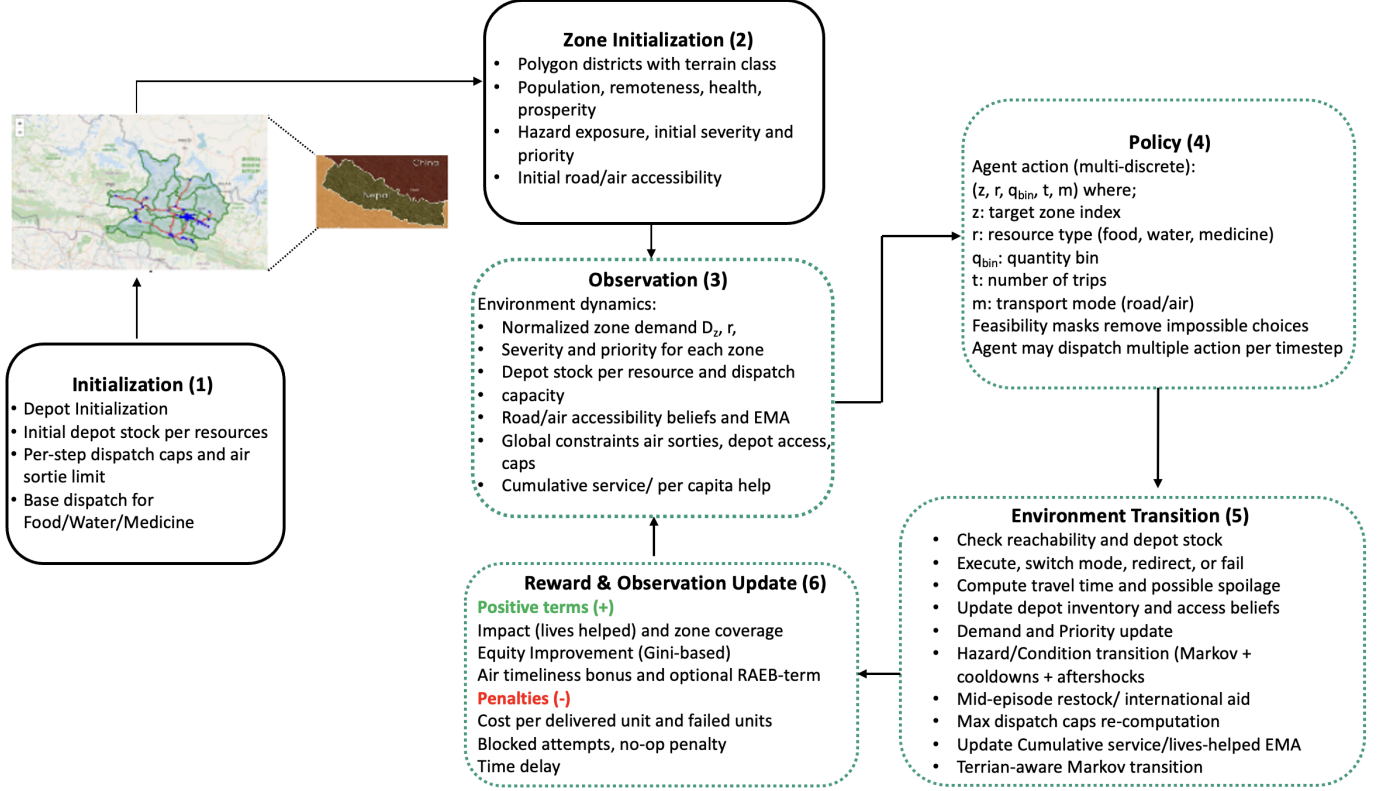


Figure 1: Overview of the disaster logistics environment. Initialization is performed once per episode, while the per-step loop follows the standard observation–action–transition–reward cycle.

where $\tau \in \{\text{plain, valley, hilly}\}$ denotes terrain type. Transition persistence is additionally regulated through stochastic cooldown intervals sampled from terrain-dependent ranges, producing longer closure durations in hilly regions than in plains.

Depot stocks are initialized in proportion to the expected “pipe capacity” of the episode and split across resources using base dispatch ratios. A mid-episode restock models international aid arrivals, with the resupply amount depending on blockage levels and average severity and ramping in over several steps.

Severity levels evolve according to a stochastic update process influenced by unmet demand, recent service history, hazard conditions, and exogenous disruptions:

$$h_{z,t+1} \sim P_h(h_{z,t+1} | h_{z,t}, c_{z,t}, \xi_{z,t}), \quad (7)$$

where $\xi_{z,t}$ denotes stochastic disruption events such as aftershocks. In practice, severity transitions are implemented using terrain-aware stochastic updates together with bounded continuous severity adjustments that are discretized into levels 1–3.

Severity updates incorporate unmet demand, recent service history, waiting time, and hazard conditions through bounded continuous adjustments with mean-reverting dynamics:

$$h_{z,t+1} = \text{clip}(h_{z,t} + \Delta h_{z,t}, 1, 3), \quad (8)$$

where $\Delta h_{z,t}$ depends on unmet demand growth, delayed service, hazard shocks, and recent relief activity.

Transport accessibility is modeled as a binary feasibility function conditioned on zone hazard states:

$$A_{z,m,t} = \begin{cases} 1, & \text{if zone } z \text{ is reachable via mode } m \text{ at time } t, \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

In addition, road accessibility depends on graph reachability under dynamic edge removals induced by blocked intermediate districts.

Accessibility is derived from the current condition state and transport mode. Under landslide conditions, road transport becomes unavailable, while combined hazards block both transport modes. During heavy rain, air accessibility may be temporarily grounded through a timestep-dependent stochastic grounding rule, while road transport remains available with reduced travel speed.

$$\Pr(A_{z,\text{Air},t} = 0) = g(c_{z,t}, \tau_z, e_z), \quad (10)$$

where $g(\cdot)$ is a stochastic grounding function increasing with adverse weather conditions, terrain difficulty, and hazard exposure.

For exposition, the simulator updates the latent state through

a sequence of conditional components:

$$P_D = P(D_{t+1} | x_t, a_t), \quad (11)$$

$$P_h = P(h_{t+1} | h_t, D_t, c_t, H_t, \xi_t), \quad (12)$$

$$P_c = P(c_{t+1} | c_t), \quad (13)$$

$$P_A = P(A_{t+1} | c_t, G_t, m, t), \quad (14)$$

$$P_{stock} = P(S_{t+1} | S_t, a_t). \quad (15)$$

These conditional components describe the sequential simulator update procedure and do not imply conditional independence between environment dynamics. This structured description mirrors the sequential simulator update process, while the full environment dynamics remain jointly coupled and stochastic in implementation.

3.1.5. Reward Signal.

The per-step reward R_t combines impact, coverage, equity, timeliness, and operational efficiency:

$$\begin{aligned} R_t = & \alpha_{\text{help}} R_{\text{help}} + \alpha_{\text{cov}} R_{\text{cov}} + \alpha_{\text{eq}} R_{\text{eq}} + \alpha_{\text{air}} R_{\text{air}} + \alpha_{\text{raeb}} R_{\text{raeb}} \\ & - (\lambda_{\text{cpu}} P_{\text{cpu}} + \lambda_{\text{fail}} P_{\text{fail}} + \lambda_{\text{block}} P_{\text{block}} \\ & + \lambda_{\text{noops}} P_{\text{noops}} + \lambda_{\text{delay}} P_{\text{delay}}). \end{aligned} \quad (16)$$

where R_{help} rewards weighted lives helped relative to a smoothed target, R_{cov} encourages broad zone coverage, R_{eq} rewards improvements in per-capita equity (via a Gini-based measure), and R_{air} provides a bonus when air deliveries materially reduce travel time for high-severity zones. An optional RAEB-style term R_{raeb} scales exploration bonuses by both route uncertainty and humanitarian need [29]. Penalties include cost per delivered unit (P_{cpu}), failed units and spoiled deliveries (P_{fail}), repeatedly probing blocked corridors (P_{block}), choosing no action when reachable unmet demand exists (P_{noops}), and accumulated time delays (P_{delay}). The coefficients are selected empirically to balance impact, fairness, feasibility, and operational efficiency, and the exact values are reported in the experimental setup. The reward formulation is intentionally multi-objective to prevent any single component from dominating the learning signal.

3.1.6. Deterministic and Stochastic Modes.

To disentangle algorithmic stability from environmental randomness, the environment can operate in deterministic or stochastic mode. Both modes share the same structural logic and reward formulation, but differ in the degree of stochasticity injected into transition dynamics. In deterministic mode, condition transitions, cooldowns, initial demand, severity, and restock schedules are computed without randomness (e.g., using argmax transitions and midpoint cooldowns), providing high reproducibility for baseline comparisons. In stochastic mode, these quantities are drawn from the underlying probabilistic models: cooldowns are sampled from annealed intervals, aftershocks occur at random steps, demand and priority include noise, and air grounding depends on sampled terrain-adjusted thresholds. This split allows us to study both best-case planning performance and robustness to realistic uncertainty.

Stochastic Transition Control. We formalize the difference between deterministic and stochastic modes through a control parameter $\kappa \in [0, 1]$ that governs transition randomness:

$$P^{(\kappa)}(x_{t+1} | x_t, a_t) = (1 - \kappa)\delta(x_{t+1} = \hat{x}_{t+1}) + \kappa\tilde{P}(x_{t+1} | x_t, a_t), \quad (17)$$

where $\hat{x}_{t+1}$ denotes the deterministic transition and $\tilde{P}$ is the stochastic transition kernel. The parameter κ provides an abstract interpolation between deterministic and stochastic transition regimes and is implemented through curriculum-based stochastic scheduling during training. Early training phases use near-deterministic transitions, while later phases progressively activate larger stochastic perturbations and probabilistic event dynamics.

3.1.7. Thailand Flood Extension

To test cross-scenario generalization, the same PO-MDP architecture is applied to a second case study inspired by the 2011 Chao Phraya basin floods in Thailand. The Thailand configuration reuses the state representation, action factorization, reward design, and training interface, but substitutes Nepal's districts with a set of flood-prone provinces selected based on historical hazard-exposure scores Fig. B.15 (b): Pathum Thani, Nonthaburi, Nakhon Sawan, Lop Buri, Sing Buri, Ang Thong, Suphan Buri, Chai Nat, Uthai Thani, Bangkok, Nakhon Pathom, and Phra Nakhon Si Ayutthaya. In place of seismic-landslide dynamics, the environment models rainfall-driven accessibility disruptions and introduces a third transport mode (boat) to capture river-based mobility

3.2. Proposed Model Architecture

Dynamic disaster logistics involves high-dimensional observations, partial observability, and uncertainty in road/air accessibility, demand evolution, and hazard transitions. To address these challenges, we design a Multi-Head actor-critic architecture based on Post-Decision Proximal Policy Optimization (PD-PPO) Actor-Critic architecture that learns *what to deliver*, *where to deliver*, and *how to deliver*, while satisfying feasibility constraints and temporal dependencies. The model integrates (i) a structured autoregressive policy, (ii) a Long Short-Term Memory (LSTM) memory backbone, (iii) feasibility-aware action masking, and (iv) an auxiliary belief module that predicts future accessibility.

System Overview. Figure 2 illustrates the overall architecture of the proposed framework. At each timestep, the environment produces an observation o_t , which is encoded into a latent representation and processed by an LSTM to capture temporal dependencies. The autoregressive actor sequentially samples structured actions under feasibility constraints, while the critic evaluates state values for policy optimization. The belief module operates only on observable trajectory information and does not access hidden simulator state. The entire system is trained using PD-PPO under curriculum-controlled environment stochasticity.

space would be intractable. We therefore adopt an autoregressive factorization:

$$\pi(a_t | o_t) = \prod_{k=1}^5 \pi_k(a_t^{(k)} | a_t^{(<k)}, h_t^{\text{LSTM}}), \quad (22)$$

where each head yields a distribution conditioned on previously sampled components. The action is generated sequentially in the order $(z \rightarrow m \rightarrow r \rightarrow t_{\text{trip}} \rightarrow q)$, reflecting the operational decision hierarchy. This autoregressive factorization substantially reduces the effective combinatorial complexity of the joint action space, enabling tractable policy learning in high-dimensional settings.

Feasibility Masking. At every head k , infeasible actions (e.g., exceeding depot stock, invalid trip count, or mode blocked by conditions) are masked by setting their logits to $-\infty$:

$$\tilde{\ell}_i^{(k)} = \begin{cases} \ell_i^{(k)}, & \text{if action is feasible,} \\ -\infty, & \text{otherwise.} \end{cases} \quad (23)$$

Feasibility Resolution and Safe Fallback: In high-dimensional constrained action spaces, it is possible for all candidate actions to become infeasible due to simultaneous violations of accessibility, capacity, or demand constraints. To ensure stable learning and prevent degenerate policy updates, we introduce a hierarchical feasibility resolution mechanism.

First, infeasible components are locally repaired by selecting the feasible alternative consistent with local masking constraints (e.g., valid trip counts or quantity bins). If no feasible dispatch exists due to extreme constraints, the environment falls back to a safe no-op behavior while preserving numerical stability.

If all sampled actions are infeasible, a global fallback mechanism is applied. This mechanism selects a minimal feasible dispatch action by scanning across zones and resources for any valid (z, m, r, t, q) combination satisfying capacity and accessibility constraints.

As a final safeguard, if no feasible dispatch exists due to extreme constraints (e.g., all routes blocked), a minimal safe action with negligible allocation is executed.

This ensures that the policy remains well-defined over all states and improve robustness under severe stochastic disruptions.

3.2.3. Accessibility Belief Module

To support anticipation of blocked or unreliable routes without leaking privileged information, we include an auxiliary belief head that predicts future accessibility:

$$\hat{p}_{t,z} = [\hat{p}_{t,z}^{\text{road}}, \hat{p}_{t,z}^{\text{air}}] = \sigma(W_b h_t^{\text{LSTM}} + b_b) \in [0, 1]^2, \quad (24)$$

representing predicted openness probabilities for road and air accessibility.

A soft-OR belief indicating overall serviceability is given by:

$$b_{t,z} = \hat{p}_{t,z}^{\text{road}} + \hat{p}_{t,z}^{\text{air}} - \hat{p}_{t,z}^{\text{road}} \hat{p}_{t,z}^{\text{air}}. \quad (25)$$

During mild-stochastic training, accessibility beliefs may additionally be smoothed using an exponential moving average (EMA) of realized route accessibility:

$$\bar{b}_{t,z} = (1 - \alpha)\bar{b}_{t-1,z} + \alpha \text{open}_{t,z}. \quad (26)$$

The belief signal is incorporated as a soft logit bias rather than a hard mask, preserving exploration while allowing downstream feasibility resolution mechanisms to remain active. The belief module therefore provides a learned approximation of latent accessibility patterns within the underlying POMDP.

3.2.4. Post-Decision Critic.

In addition to the standard state critic $V_\phi(o_t)$, the proposed framework incorporates a post-decision critic $V_\phi^{PD}(o_t, a_t)$ [13, 14] that estimates the expected value immediately after action execution but before stochastic environment transitions fully unfold.

$$x_t^{PD} = f_{\text{exec}}(x_t, a_t), \quad (27)$$

where the immediate effects of the selected dispatch action have been applied, but subsequent stochastic environment transitions have not yet been sampled.

$$V_\phi^{PD}(o_t, a_t) \approx \mathbb{E}[R_{t:\infty} | x_t^{PD}]. \quad (28)$$

The post-decision critic is incorporated into advantage and return estimation through an auxiliary TD blending mechanism that stabilizes temporal-difference targets under stochastic disruptions, delayed accessibility changes, and non-stationary demand evolution. The standard critic $V_\phi(o_t)$ is retained to provide a conventional state-value baseline within the proposed PD-PPO optimization framework. The contribution of the post-decision critic is additionally evaluated through ablation experiments in Section 4.

3.3. Model Training

The proposed actor-critic agent is trained with a PD-PPO variant tailored to the stochastic, partially observable disaster environment. Training combines on-policy rollouts, advantage estimation, bounded reward shaping, curriculum learning, and trust-region style regularization to obtain stable policies under high uncertainty [12, 28, 34]. While the optimization objective follows standard PPO, the proposed training procedure differs in three key aspects: (i) structured autoregressive action sampling under feasibility constraints, (ii) belief-informed representations for partially observable dynamics, and (iii) curriculum-controlled stochasticity to stabilize learning under non-stationary conditions.

3.3.1. Rollout Generation and Policy Update

Training is performed in an on-policy manner with N parallel copies of the environment. At each update cycle, the agent interacts with the environment for T time steps, generating trajectories $\{(o_t, a_t, r_t, o_{t+1})\}_{t=0}^{T-1}$, where o_t is the observation at time t , a_t is the sampled multi-head action tuple, and r_t is the scalar reward. Action sampling is performed autoregressively under feasibility constraints. At each decision head, infeasible actions

are masked out, and valid components are sampled sequentially. Sampled actions are then repaired or filtered to satisfy capacity and accessibility constraints. If no valid actions remain, a fallback dispatch is selected to keep the rollout well-defined.

For each timestep, the rollout buffer stores tuples of the form

$$(o_t, a_t, r_t, V_\phi(o_t), \log \pi_\theta(a_t|o_t), d_t),$$

where d_t denotes the terminal indicator. These quantities are used to compute Generalized Advantage Estimation (GAE) [35]:

$$\delta_t = r_t + \gamma V_\phi(o_{t+1}) - V_\phi(o_t), \quad (29)$$

$$\hat{A}_t = \delta_t + \gamma \lambda_t \hat{A}_{t+1}. \quad (30)$$

The post-decision critic $V_\phi^{PD}(o_t, a_t)$ is incorporated as an auxiliary stabilization signal through a blended return target:

$$\tilde{G}_t = (1 - \eta)(\hat{A}_t + V_\phi(o_t)) + \eta V_\phi^{PD}(o_t, a_t), \quad (31)$$

where $\eta \in [0, 1]$ controls the contribution of the auxiliary post-decision estimate under increasing stochasticity. Advantages are then normalized using a median absolute deviation (MAD) scheme to improve robustness to outliers:

$$\tilde{A}_t = \frac{\hat{A}_t - \text{median}(\hat{A})}{\text{MAD}(\hat{A}) + \varepsilon}. \quad (32)$$

followed by clipping to a bounded range. This reduces sensitivity to outliers arising from stochastic transitions. MAD normalization was adopted instead of standard variance normalization to improve robustness against large reward fluctuations arising from stochastic disruptions and rare failure events.

PPO updates the policy by maximizing the clipped surrogate objective [12]:

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min(r_t(\theta) \tilde{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \tilde{A}_t) \right], \quad (33)$$

where $r_t(\theta) = \pi_\theta(a_t | o_t) / \pi_{\theta_{\text{old}}}(a_t | o_t)$ is the importance ratio and ϵ is the clipping parameter. The state critic parameters ϕ and the auxiliary post-decision critic are trained using temporal-difference regression, while policy optimization additionally incorporates Kullback–Leibler (KL)-regularized stabilization; see Algorithm 1). An entropy bonus encourages exploration in the early curriculum phases. All expectations are taken over trajectories generated under the curriculum-controlled environment dynamics described in Section 3.1.6.

3.3.2. Reward Shaping

The environment-level reward defined in Section 3.1.5 decomposes mission performance into impact (lives helped), coverage, equity, cost efficiency, timeliness, and feasibility penalties. During training, auxiliary shaping terms such as transport cost, delay, blocked attempts, and light RAEB-style exploration bonus are bounded and scaled so that the primary humanitarian objectives dominate. This improves credit assignment and learning stability under stochastic transitions.

3.3.3. Curriculum Learning

To stabilize learning in a highly stochastic environment, we employ a three-phase curriculum [32, 34] over environment complexity:

1. **Warm stationary.** Deterministic dynamics, road-only transport, no mid-episode restock or aftershocks. This phase allows the agent to learn basic dispatch patterns without disruptive events.
2. **Mild stochastic.** Moderate randomness in demand, conditions, and accessibility, with limited aftershocks and restock events. soft accessibility estimates derived from EMA-based beliefs are incorporated into policy conditioning.
3. **Full stochastic.** Full environment complexity is enabled, including terrain-aware Markov condition transitions, stochastic aftershocks, dynamic restocking, and full reward shaping.

The curriculum progressively changes environment settings, including stochastic accessibility, aftershocks, restocking, masking rules, and progressively relaxed environment constraints, thereby increasing the effective transition uncertainty over training. Formally, we define a schedule:

$$\kappa(t) = \sigma(\alpha(t - t_0)), \quad (34)$$

where $\sigma(\cdot)$ is a sigmoid scheduling function controlling the progression from deterministic to fully stochastic training phases.

3.3.4. Regularization and Stability Control

Several mechanisms are employed to control variance and stabilize actor–critic training. Feasibility masks derived from environment constraints (e.g., stock limits, quantity caps, and trip budgets) set logits of infeasible actions to $-\infty$, ensuring that probability mass is assigned only to valid decisions (cf. Section 3.2.2).

An entropy regularization term is applied to the joint autoregressive policy, computed as the sum of entropies across all action heads:

$$\mathcal{L}_{\text{entropy}} = -\beta_t \mathbb{E}_t [\mathcal{H}(\pi_\theta(\cdot|o_t))], \quad (35)$$

where β_t decreases over training.

Gradient norms are clipped to prevent unstable parameter updates. Additionally, KL-based policy divergence monitoring is used to track the deviation between successive policies:

$$\text{KL}(\pi_{\theta_{\text{old}}}(\cdot|o_t) \parallel \pi_\theta(\cdot|o_t)) \leq \delta, \quad (36)$$

where δ is a target threshold. In addition, a soft KL penalty is incorporated into the objective:

$$\mathcal{L}_{\text{KL}} = \beta \text{KL}(\pi_{\theta_{\text{old}}}(\cdot|o_t) \parallel \pi_\theta(\cdot|o_t)), \quad (37)$$

where β is adaptively increased when the KL divergence exceeds a target threshold.

Algorithm 1: Stabilized Autoregressive PD-PPO with Feasibility Enforcement for Stochastic Disaster Logistics

Input: Environment $\mathcal{E}$, policy π_θ , value function V_ϕ , post-decision critic V_ϕ^{PD} , curriculum schedule $\kappa(t)$
Initialize: parameters θ, ϕ
for iteration = 1 to N **do**
 Set $\kappa \leftarrow \kappa(\text{iteration})$
 Initialize buffer $\mathcal{B}$
 for episode = 1 to M **do**
 Reset o_0
 for $t = 0$ to $T - 1$ **do**
 Observe o_t
 Encode h_t
 Update h_t^{LSTM}
 // Autoregressive sampling + repair + fallback
 Sample a_t
 Apply feasibility repair
 Remove infeasible actions
 if no valid action **then**
 Apply fallback action
 Execute a_t , observe r_t
 Store transition in $\mathcal{B}$
 if terminal **then**
 break
 // GAE with post-decision critic
 Compute post-decision TD targets and advantages
 ; // MAD normalization
 Normalize $\hat{A}_t$
 // TD-blended return
 Compute R_t
 for $k = 1$ to K **do**
 Sample minibatch
 Compute $r_t(\theta)$
 Compute $\mathcal{L}^{\text{PPO}}$
 Compute $\mathcal{L}_V$
 Compute $\mathcal{L}_{\text{PD}}$
 Compute entropy + KL penalty
 Update θ, ϕ
Output: π_θ^*

For the post-decision critic variant, we employ a dual-critic structure consisting of a state-value function $V_\phi(o_t)$ and a post-decision value function $V_\phi^{\text{PD}}(o_t, a_t)$, trained using a smooth- L_1 (Huber) loss:

$$\mathcal{L}_{\text{PD}} = \mathbb{E}_t \left[\text{Huber}(V_\phi^{\text{PD}}(o_t, a_t) - (r_t + \gamma V_\phi(o_{t+1}))) \right]. \quad (38)$$

The state-value function is trained using a regression loss:

$$\mathcal{L}_V = \mathbb{E}_t \left[\left(V_\phi(o_t) - R_t \right)^2 \right] \quad (39)$$

3.3.5. PD-PPO Training Algorithm with Stabilization Components

Algorithm 1 summarizes the training procedure used in our framework. The method is based on Proximal Policy Optimization (PPO) and incorporates key stabilization components, including autoregressive action decomposition, feasibility-aware masking, and curriculum-controlled stochasticity. Optional components such as the post-decision critic and belief-based conditioning are included for ablation analysis. An entropy

regularization term and KL penalty are added to stabilize training. The KL controller is used only as an auxiliary stabilization mechanism and does not replace the clipped PPO objective.

4. Results and Analysis

This section evaluates the proposed PD-PPO-based framework in the MultiDistrictDisasterEnv under both deterministic and stochastic settings. We focus on four key aspects: (i) the ability of the framework to coordinate multi-zone resource allocation under partial observability and dynamic constraints; (ii) its effectiveness in handling delayed rewards and long-horizon dependencies arising from evolving demand, severity, and accessibility; (iii) the impact of stabilization components, including curriculum-driven stochasticity, feasibility-aware masking, recurrent state modeling, and belief-guided conditioning, on training stability and credit assignment; and (iv) the extent to which the learned policy balances efficiency, equity, and timeliness relative to non-learning baselines. Unless otherwise stated, results are reported for the Nepal configuration with $Z = 12$ districts and $R = 3$ resources. Hyperparameter and curriculum schedules follow the settings summarized in Table 1.

4.1. Training Dynamics and Stability

The policy network consists of a two-layer encoder, a three-layer LSTM recurrent module, autoregressive action heads, an accessibility belief module, and dual critic networks. Detailed architectural specifications and initialization settings are provided in Appendix H.6. Figure 3 summarizes the evolution of key training signals over 1200 iterations. The upper-left panel shows the state and post-decision critic losses. Both exhibit high variance in the warm deterministic phase (iterations 0–300), reflecting rapid adjustment to basic dispatch dynamics.

Table 1: Core training configuration for the proposed PD-PPO agent.

Setting	Value
Algorithm	PD-PPO with dual critics
Training iterations	1200 total; 300 warmup
Rollout length	512 (warm), 384 (mild), 256 (full)
Parallel actions	3 dispatch rows per step
Discount factor γ	0.99
GAE parameter λ_t	Adaptive in [0.82, 0.95]
PPO clipping ϵ	0.2
Entropy coefficient	0.03 annealed to [0.01, 0.06]
KL target	0.03 (warm/mild), 0.05 (full)
Optimization epochs	4 per PPO update
Optimizer	AdamW
Learning rates	3×10^{-4} (policy/critics), 1×10^{-4} (encoder/LSTM/conditioners), 2×10^{-4} (belief head)
Weight decay	0.01; disabled for LSTM, normalization, and bias parameters
Optimizer: AdamW optimizer	$\beta = (0.9, 0.999)$, $\epsilon = 10^{-5}$
Gradient clipping	Max norm = 0.4
Learning Rate scheduler	MultiStepLR (milestone = 700, decay = 0.5)
Curriculum	Warm $\rightarrow$ Mild stochastic $\rightarrow$ Full stochastic
Masking scheme	Hard-truth (warm), minimal-sane (main training)
Advantage normalization	MAD-based normalization

As the curriculum transitions to mild and then full stochasticity, loss magnitudes become more stable and exhibit reduced oscillatory behavior, indicating more consistent value estimation and reduced temporal-difference error. The transition points between curriculum phases are visible as short-term perturbations in entropy, critic loss, and policy loss, after which the training signals progressively re-stabilize without evidence of divergence. This behavior suggests that gradual stochastic curriculum scheduling facilitates adaptation to increasing environment uncertainty while maintaining stable optimization dynamics. As expected, diagnostic variance increases during the transition to the fully stochastic curriculum phase, particularly in temporal-difference error statistics. This reflects increased transition uncertainty arising from stochastic accessibility changes, aftershock events, and evolving demand dynamics rather than optimization instability, as policy updates remain bounded and task-level performance remains consistent across independent seeds.

The entropy trajectory (upper-right) follows the curriculum schedule. Entropy remains high (≈ 8.5 nats) during the warm phase, promoting broad exploration in a simplified environment. With increasing stochasticity and operational complexity, entropy decays to ≈ 6.5 and stabilizes, while short spikes at phase boundaries mark controlled exploration boosts during regime changes.

Episodic returns (lower-left) are highest in the deterministic warm phase, then drop when stochastic demand, blockages, and aftershocks are enabled. Returns subsequently stabilize within a comparatively narrower range, suggesting that the policy adapts to increased stochasticity without exhibiting divergent learning behavior. Policy loss (lower-right) decreases over time, with occasional peaks aligned with curriculum transitions and KL-based trust-region corrections. Overall, the training curves suggest that the combined use of curriculum-controlled stochasticity, feasibility-aware masking, recurrent memory, and adaptive normalization contributes to stable policy optimization under progressively harder operating conditions.

Table 2: Multi-seed evaluation results of the proposed PD-PPO framework under stochastic disaster conditions. Reward Mean (raw episodic return)

Metric	Seed 11	Seed 22	Seed 33	Seed 44	Seed 55
Reward Mean	38.732	38.200	38.055	38.251	37.419
Reward Std	0.281	0.134	0.334	0.348	0.215
Help Mean	25.305	25.730	25.328	24.365	24.191
Cost Mean	4.617	4.887	4.649	4.390	4.487
Block Mean	0.550	1.103	0.953	0.231	0.462
Delay Mean	0.224	0.167	0.156	0.231	0.336

To further evaluate robustness under stochastic operating conditions, additional multi-seed experiments were conducted using five independent training seeds and five fixed evaluation seeds, as summarized in Table 2. Each statistical experiment was trained for 600 iterations using the same hyperparameter configuration and curriculum schedule as the primary 1200-iteration training run. The shorter horizon was selected to balance computational cost with reproducibility analysis across multiple independent runs.

Across all seeds, the proposed PD-PPO framework demonstrated relatively stable performance, achieving reward means

between approximately 37.4 and 38.7 with comparatively low inter-evaluation variance. The results also highlight the trade-off characteristics learned by the policy. For example, Seed 22 achieved the highest humanitarian assistance score (R_{help}), but incurred comparatively higher blockage penalties, whereas Seed 44 produced the lowest blockage penalties with slightly reduced assistance performance. Such behavior suggests that the proposed framework is capable of learning diverse yet operationally feasible dispatch strategies under stochastic disaster conditions.

The relatively low reward standard deviations across evaluation seeds indicate that the learned policies generalize consistently across different stochastic realizations of demand growth, infrastructure disruption, and accessibility changes. Overall, the multi-seed experiments support the stability and robustness of the proposed curriculum-guided PD-PPO framework for dynamic humanitarian logistics optimization.

4.2. Internal Diagnostics and Value Approximation

Internal diagnostics in the lower row provides of Fig. 3 additional evidence of stable optimization. The advantage statistics show that the mean advantage oscillates around zero after the initial iterations, while the standard deviation decreases and stabilizes after ~ 400 iterations. This confirms that MAD-based advantage normalization effectively controls variance in the presence of a non-stationary reward signal.

Return statistics indicate that both the mean and spread of discounted returns increase during early learning and plateau thereafter, consistent with improving long-term performance followed by consolidation. The PD-PPO trust-region diagnostics show that empirical KL divergence remains within the desired operating range (0.03–0.05), and that the mean importance ratio and its 5th/95th percentiles remain close to 1.0 and within the [0.8, 1.2] interval. This demonstrates that the clipped surrogate objective successfully bounds policy updates and prevents destructive steps.

Running mean and standard deviation of returns (Fig. 3f) converge smoothly, indicating that running return scaling delivers a stable signal for both policy and critic updates. The post-decision critic achieves an R^2 of approximately 0.58 against realized returns (Fig. 3h), indicating that the critic captures a substantial portion of return variability despite the highly stochastic environment dynamics. While the value approximation is imperfect, PD-PPO optimization primarily requires reliable relative advantage estimation rather than exact return prediction, making this level of value estimation sufficient for stable policy improvement. The residual regression-to-the-mean behavior at extreme returns is expected in highly stochastic environments and does not prevent stable advantage estimation.

4.3. Baseline Comparison using Deterministic Policy Decoding

We compare the proposed PD-PPO policy against two non-learning baselines: (i) a Rolling Linear Programming (Rolling-LP) policy that solves a receding-horizon mixed-integer optimization problem at each decision step, and (ii) a Greedy

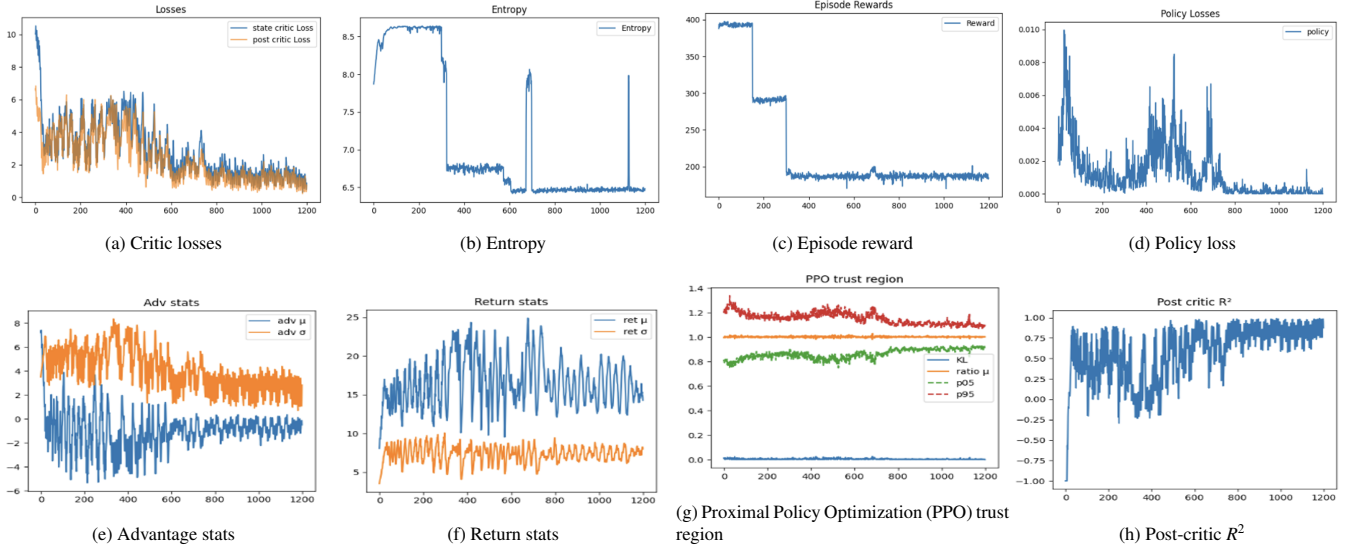


Figure 3: Training dynamics and stability diagnostics of the PD-PPO agent over 1,200 iterations. Each panel reports a different diagnostic used to assess convergence and variance control.

heuristic that ranks feasible dispatches using a local benefit-cost scoring rule Appendix C. To ensure a fair comparison between reinforcement learning and optimization-based methods, PD-PPO is evaluated using deterministic greedy policy decoding (maximum-probability action selection) rather than stochastic policy sampling.

Specifically, deployment actions are selected using the maximum-probability action from the learned policy distribution instead of sampling from the policy during evaluation. This distinction is important because stochastic action sampling primarily reflects exploration behavior used during training, whereas Rolling-LP and Greedy are inherently deterministic decision procedures. The evaluation protocol therefore isolates deployment-policy quality from exploration-induced action variance. Unless otherwise specified, environment stochasticity remains active during evaluation, including stochastic hazard evolution, accessibility disruptions, aftershock events, and demand fluctuations.

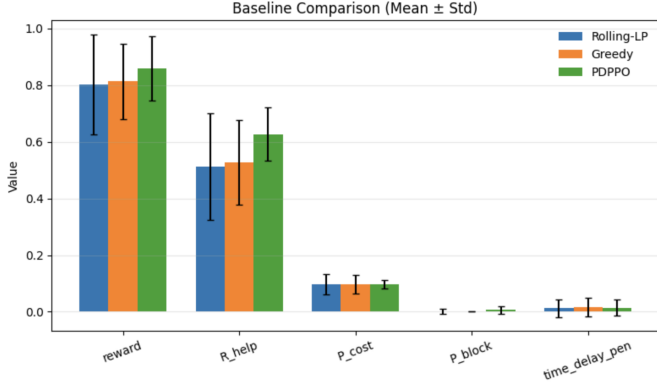
Each method was evaluated over 10 paired environment seeds with 50 episodes per seed, resulting in 500 evaluation episodes per method. Paired evaluation ensures that all methods experience identical stochastic environment realizations, including demand growth, accessibility changes, resource depletion, and environmental disruptions. Table 3 and Figs. 4b summarize the experimental results. Statistical significance analysis using paired t -tests, Wilcoxon signed-rank tests, bootstrap confidence intervals, and effect-size measurements further confirms the robustness of the observed performance differences Appendix E.

Among all methods, PD-PPO demonstrates improved delivered-help performance (0.860 ± 0.113) and delivered-help score (0.627 ± 0.094) in this deterministic policy evaluation using greedy decoding under stochastic environment dynamics. Importantly, these gains are achieved while maintaining operational costs comparable to the optimization-based methods.

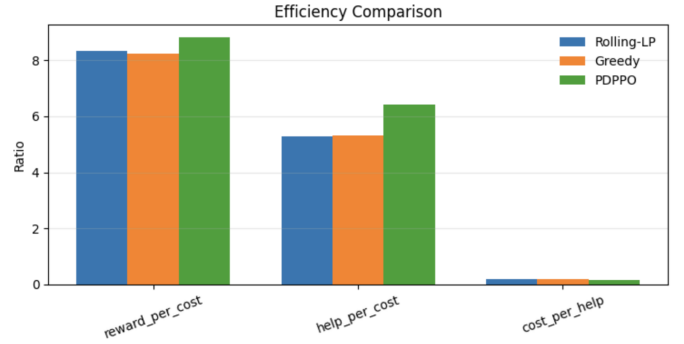
Although PD-PPO incurs slightly higher blockage penalties than the deterministic optimization baselines, it simultaneously achieves the lowest delay penalty, indicating that the learned policy prioritizes timely and adaptive dispatching under uncertainty.

Rolling-LP remains highly competitive in terms of cumulative reward while maintaining low operational cost. However, its considerably larger variance across evaluation episodes suggests greater sensitivity to stochastic disruptions such as changing accessibility, demand growth, and evolving resource constraints. The Greedy heuristic exhibits similar cost characteristics but delivers less total assistance and demonstrates reduced adaptability under stochastic dynamics. The Pareto analysis in Fig. 4c further highlights the trade-off between operational cost and humanitarian assistance. The Pareto visualization focuses on the two primary competing reward components: humanitarian assistance (R_{help}) and operational transport cost (P_{cost}), capturing the trade-off between aid delivery and resource expenditure.

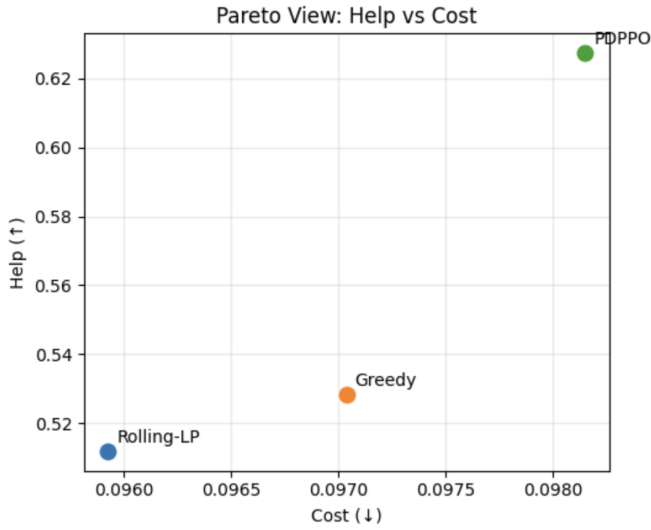
The reward-distribution boxplots in Fig. 4d provide additional insight into policy robustness. PD-PPO exhibits the highest median episode reward together with consistently lower reward dispersion across stochastic realizations. In contrast, Rolling-LP demonstrates consistent larger dispersion and more extreme low-reward outcomes, suggesting higher vulnerability to environmental uncertainty despite strong average-case optimization performance. Although PD-PPO exhibits several low-reward outlier episodes, the majority of trajectories remain tightly concentrated around a higher median reward compared to the optimization baselines. These outliers typically arise from rare stochastic combinations of accessibility disruptions, delayed replenishment, and sequential dispatch commitment errors that compound over long horizons. Overall, the results indicate that experience-driven sequential policies can achieve both higher expected utility and improved operational stability



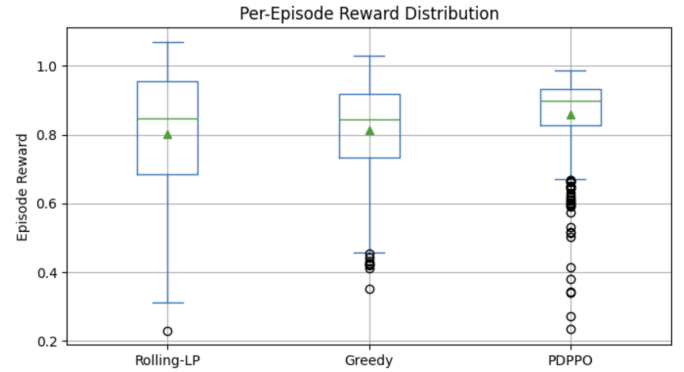
(a) Deterministic: core metrics (mean $\pm$ sd)



(b) Stochastic: core metrics (mean $\pm$ sd)



(c) Deterministic: Pareto view (help vs. cost)



(d) Deterministic Reward Distribution

Figure 4: Comparison of Rolling-Linear Programming, Greedy, and PD-PPO across deterministic evaluation metrics, efficiency trade-offs, Pareto cost-help analysis, and reward stability. PD-PPO achieves higher delivered help and more stable performance with competitive operational cost.

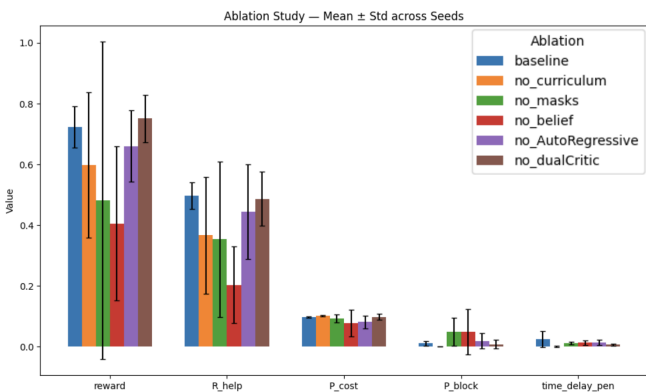


Figure 5: Ablation study: mean $\pm$ standard deviation across seeds for reward, help delivered, and penalty metrics.

in highly dynamic disaster-response environments.

A separate scalability experiment varies the number of zones and resources, reusing a policy trained on the full ($Z=12, R=3$) environment as summarized in Table Appendix A.1. PD-PPO maintains consistent normalized reward across smaller problem

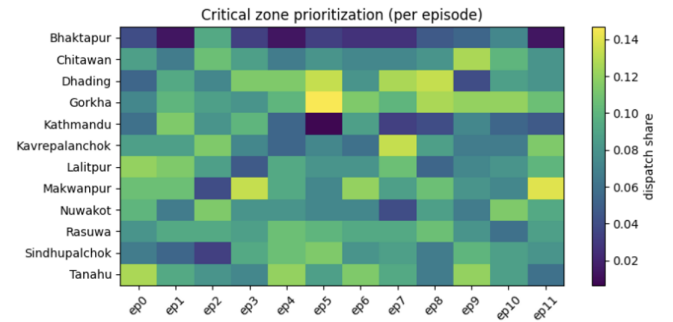


Figure 6: Critical zone prioritization across episodes

instances at approximately 0.7–0.8, while inference remains approximately 20–40 $\times$ faster than repeatedly solving Rolling-LP optimization problems. This highlights the potential of trained policies as computationally efficient surrogates for optimization in time-critical humanitarian operations.

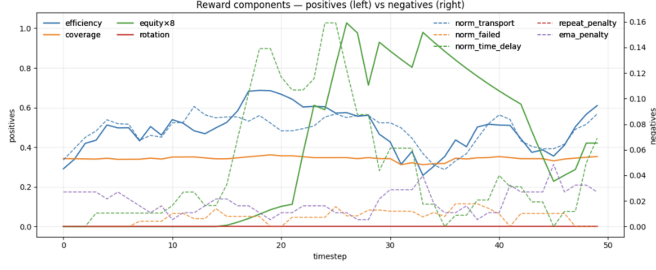


Figure 7: Reward Decomposition

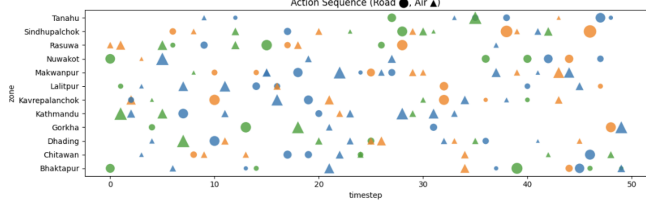


Figure 8: Action sequence (road vs. air, resources)

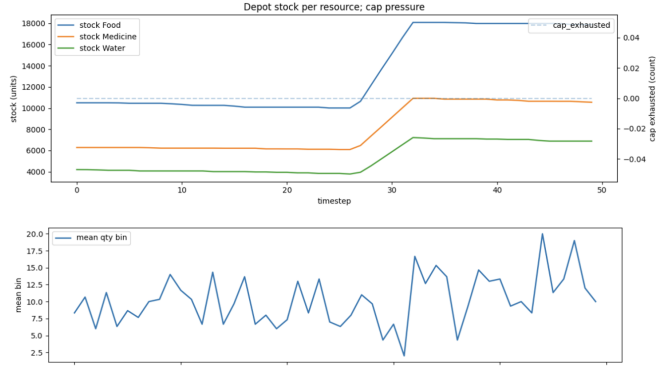


Figure 9: Inventory and Resource Dispatch

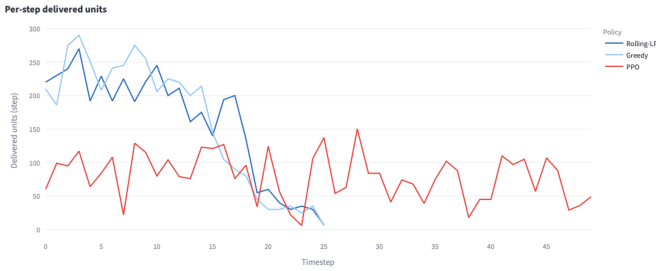


Figure 10: Long term efficiency and credit assignment

Table 3: Performance comparison under deterministic policy evaluation across 10 unseen environment seeds (500 paired rollout episodes in total). Values are reported as mean $\pm$ standard deviation.

Metric	Rolling-LP	Greedy	PD-PPO
Reward	0.802 $\pm$ 0.177	0.813 $\pm$ 0.134	0.860 $\pm$ 0.113
R_{help}	0.512 $\pm$ 0.188	0.528 $\pm$ 0.149	0.627 $\pm$ 0.094
P_{cost}	0.096 $\pm$ 0.036	0.097 $\pm$ 0.034	0.098 $\pm$ 0.015
P_{block}	0.001 $\pm$ 0.008	0.000 $\pm$ 0.000	0.006 $\pm$ 0.014
P_{delay}	0.012 $\pm$ 0.031	0.017 $\pm$ 0.033	0.014 $\pm$ 0.028

Additional stress-testing under low-inventory conditions (Fig. 10), where the initial depot stock was capped at 20%, shows that both Rolling-LP and Greedy rapidly exhaust avail-

able inventory due to myopic stepwise allocation behavior. In contrast, PD-PPO maintains broader service coverage throughout a larger portion of the episode by leveraging longer-horizon credit assignment and more anticipatory dispatch behavior.

From an operational perspective, reduced performance variance is particularly important in disaster-response settings, where highly unstable policies may lead to severe under-service during adverse stochastic realizations. Compared with Rolling-LP and Greedy baselines, PD-PPO exhibits tighter reward dispersion and more concentrated confidence bounds across evaluation seeds, indicating improved robustness and more predictable deployment behavior under uncertainty.

Lastly, although value-based reinforcement learning methods such as Q-learning and Deep Q-Networks (DQN) are widely used in sequential decision problems, direct application in the present setting is challenging due to the combinatorial structure of the dispatch action space.

A single dispatch action consists of sequential selections over zone, transport mode, resource type, trip count, and quantity allocation across multiple parallel dispatch rows:

$$a = (z, m, r, t, q)_{1:K}.$$

Under joint action enumeration, the effective action-space cardinality grows approximately as

$$|\mathcal{A}| \approx (Z \cdot M \cdot R \cdot T \cdot Q)^K,$$

where Z , M , R , T , and Q denote the numbers of zones, transport modes, resource types, trip levels, and quantity bins, respectively, and K represents the number of parallel dispatch rows. Even moderate problem instances therefore produce extremely large discrete action spaces.

Value-based methods additionally require Bellman maximization over the joint action space,

$$\max_{a'} Q(s', a'),$$

during temporal-difference updates. In the present environment, this optimization becomes particularly challenging because the feasible action subset varies dynamically with inventory availability, accessibility disruptions, dispatch constraints, and stochastic hazard evolution. Consequently,

$$\mathcal{A}_{\text{valid}}(s) \subset \mathcal{A}, \quad |\mathcal{A}_{\text{valid}}(s)| \ll |\mathcal{A}|,$$

resulting in highly sparse and state-dependent feasible-action regions.

In contrast, autoregressive policy-gradient methods naturally support conditional action decomposition through sequential factorization,

$$\pi(a|s) = \prod_k \pi(a_k | a_{<k}, s),$$

allowing structured dispatch decisions to be generated through a series of lower-dimensional conditional policies rather than exhaustive joint action evaluation. This factorization aligns directly with the operational structure of the humanitarian dispatch process considered in this work and avoids the need for

explicit combinatorial Bellman maximization over the full joint action space. Accordingly, the experimental evaluation focuses on comparisons against optimization-based and heuristic dispatch baselines operating under the same environmental constraints.

4.4. Ablation Study

To examine the contribution of key architectural and training components, we perform an ablation study in which each variant is retrained from scratch using three independent stochastic seeds. Fig. 5 reports the mean and standard deviation of normalized reward, delivered help, transport cost, blockage penalties, and delay penalties across configurations. Detailed confidence intervals and paired comparisons relative to the baseline policy are provided in Table Appendix D.3.

The baseline configuration combines curriculum scheduling, feasibility-aware masking, autoregressive policy factorization, the accessibility belief module, and the dual-critic PD-PPO training structure. Each ablation removes one component while keeping the remaining architecture unchanged.

Removing curriculum scheduling is associated with somewhat lower delivered help together with higher blockage and delay penalties. However, the confidence intervals for total reward and delivered help partially overlap with the baseline policy, suggesting that the main effect of curriculum scheduling is improved performance consistency and variance control across stochastic training conditions rather than a large change in final episodic performance.

Disabling feasibility-aware masking (no masks) produces the largest increase in variance across seeds and generally lower delivered help. Without structured feasibility filtering, the policy more frequently samples operationally inefficient dispatch combinations, making optimization less stable in the large combinatorial action space. In this setting, sampled actions are repaired through deterministic feasibility projection before execution so that environment interaction remains valid while isolating the effect of policy-level masking.

Removing the accessibility belief module (no belief) is associated with lower average reward and delivered help relative to the baseline policy. The paired confidence intervals indicate a more consistent reduction in performance compared with the other ablations, suggesting that belief-guided accessibility estimation provides useful context under partial observability and stochastic hazard dynamics.

The autoregressive ablation (no autoregressive) also shows moderate reductions in reward and delivered help together with slightly increased cost variability. Although the observed differences remain modest, the results suggest that sequential conditional action factorization can improve coordination among zone, transport, resource, and quantity decisions.

Finally, removing the post-decision critic (no dual critic) results in comparatively small changes in reward and delivered help, with substantial overlap in confidence intervals. This suggests that the auxiliary post-decision critic primarily contributes to optimization robustness and temporal credit assignment during training rather than directly improving final episodic performance. Although the single-critic

variant occasionally achieves slightly higher average reward, it exhibits greater variability across seeds, indicating that the post-decision critic mainly acts as a stabilizing component under stochastic environment dynamics. Supporting optimization diagnostics are provided in Appendix F.

Overall, the ablation results suggest that feasibility-aware masking and accessibility belief estimation provide the largest direct contribution to reward quality and humanitarian assistance. In contrast, curriculum scheduling, autoregressive factorization, and the post-decision critic appear to improve robustness, coordination consistency, and variance control under stochastic environment dynamics, although their direct impact on average reward is comparatively smaller.

4.5. Reward Sensitivity

To examine the influence of reward composition on learned dispatch behavior, we performed a reward sensitivity analysis under multiple operational weighting profiles, including efficiency-oriented, equity-oriented, humanitarian, and robustness-focused settings. The default reward configuration combines delivered help, coverage, deprivation-equity balancing, operational cost, failure penalties, blockage penalties, and no-op penalties. The corresponding default coefficients are:

$$\begin{aligned} w_{\text{help}} &= 0.75, & w_{\text{cov}} &= 0.60, & w_{\text{deq}} &= 0.40, \\ w_{\text{cost}} &= 0.10, & w_{\text{fail}} &= 0.20, & w_{\text{block}} &= 0.06, \\ w_{\text{noops}} &= 0.18. \end{aligned} \quad (40)$$

Table 4 summarizes the modified reward coefficients relative to this default configuration. Each reward profile was evaluated across three independent stochastic seeds, while detailed confidence intervals and paired statistical comparisons are reported in Appendix D.1 and Appendix D.2. Fig. 11 summarizes the resulting trade-offs across normalized reward, aid reach, operational cost, blockage penalties, and delay. The efficiency-oriented profile achieves lower operational cost and delay penalties, but also produces lower delivered help, indicating that stronger cost minimization reduces overall service coverage.

Table 4: Reward-weight profiles used in sensitivity analysis. Only modified weights relative to the default configuration are shown.

Profile	Modified reward weights
Efficiency	$w_{\text{cost}}=0.35, w_{\text{deq}}=0.10$
Equity	$w_{\text{deq}}=1.00, w_{\text{cost}}=0.02$
Humanitarian	$w_{\text{help}}=1.25, w_{\text{cov}}=0.75,$ $w_{\text{fail}}=0.50, w_{\text{block}}=0.50, w_{\text{cost}}=0.05$
Robustness	$w_{\text{block}}=0.80, w_{\text{noops}}=0.20, w_{\text{cost}}=0.10$

In contrast, the humanitarian configuration places greater emphasis on delivered assistance and coverage, resulting in higher aid reach together with increased operational penalties. Increasing the deprivation-equity coefficient produces a more balanced allocation behavior, maintaining moderate delivered help while avoiding large increases in operational cost. The robustness-focused profile reduces blockage-related penalties while maintaining comparatively stable delivered assistance under stochastic disruptions, suggesting that the policy adapts to-

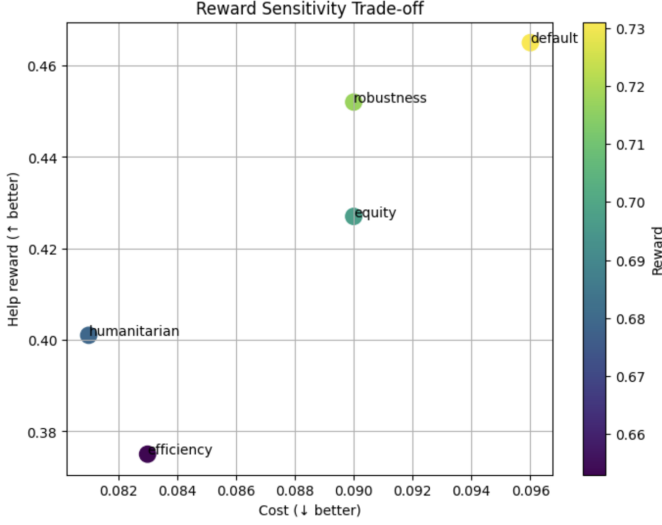


Figure 11: Reward sensitivity trade-offs under different reward-weighting profiles. Marker color indicates normalized episodic reward.

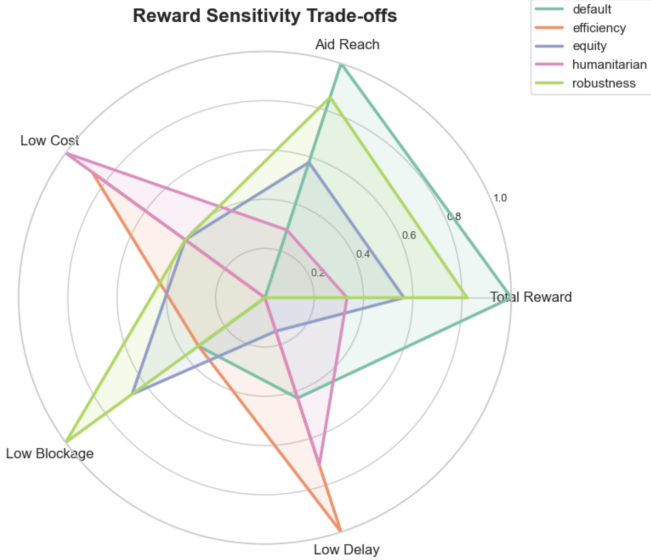


Figure 12: Radar visualization of normalized operational metrics under different reward-weighting profiles.

ward more conservative dispatch behavior when accessibility penalties are emphasized.

Overall, the results indicate that the proposed framework responds consistently to changes in reward priorities and can adapt its dispatch strategy toward different operational objectives. The accompanying Pareto and radar visualizations therefore provide an interpretable projection of the broader multi-objective reward landscape rather than a complete representation of all reward components.

4.6. Policy Interpretability and Explainability

To assess the realism and transparency of the learned policy, we analyze episode-level behavior using reward decomposition and spatial-temporal dispatch visualizations (Fig. 7).

Reward attribution plots indicate that positive reward contributions are primarily associated with efficiency and cover-

age terms, while equity-related contributions increase during the mid-episode phase as previously under-served districts receive additional aid. Transport and delay penalties remain low, while failure and blockage penalties appear as occasional spikes aligned with accessibility disruptions. Overall, the positive reward components generally remain larger than the aggregated penalty terms across most time steps, suggesting stable operational behavior despite stochastic disruptions.

(Fig. 6) shows how dispatch priorities shift across episodes as district conditions evolve. The dispatch heatmaps suggest that the PD-PPO policy reallocates attention across zones in response to changing severity, unmet demand, priority, and accessibility conditions. This episode-wise redistribution suggests that the learned policy adapts its dispatch allocation according to evolving operational conditions and accessibility constraints while partially reflecting the deprivation-aware reward structure. Road transport appears more frequently in closer and better-connected districts, whereas air dispatch is more commonly selected for remote or mountainous regions where road accessibility is more uncertain (Fig. 8).

Resource-quantity plots (Fig. 9) show that the agent increases shipment volumes following depot restocks and scales down quantities when stock becomes scarce, suggesting inventory-aware dispatch behavior. These qualitative patterns are broadly consistent with expected humanitarian logistics behavior, suggesting that the learned policy remains operationally interpretable under stochastic disaster conditions.

Table 5: Representative stochastic rollout decisions of the proposed PD-PPO policy. Rewards denote aggregate timestep-level outcomes after executing all dispatch actions.

t	Observation	Action	Trips	Feasibility	Outcome	Reward	Notes
4	Z1 sev=2, need=0.35	Road R2 q=4	3	Yes	Fail	+1.04	other
10	Z11 sev=3, need=0.52	Air R2 q=11	1	Yes	Success	+1.09	high need
22	Z5 sev=2, need=0.39	Air R0 q=20	2	Yes	Success	+0.73	-
26	Z4 sev=1, need=0.20	Road R0 q=2	4	Yes	Fail	+0.87	cap_exhaus
39	Z11 sev=2, need=0.52	Road R1 q=18	1	Yes	Success	+0.99	-
47	Z2 sev=2, need=0.41	Road R2 q=12	1	Yes	Success	+0.99	-

Abbreviations: t = Decision timestep; Z = Zone; sev = Severity level; need = Normalized humanitarian need score; R = Resource type (R0 = Food, R1 = Water, R2 = Medicine); q = Quantity bin.

At Table 5 the selected examples illustrate the policy’s ability to dynamically allocate heterogeneous resources using both road and air transportation modes while operating under uncertain environmental conditions and feasibility constraints.

Several observations can be made from the rollout trajectory. First, the policy demonstrates adaptive multi-modal behavior by alternating between road and air dispatches depending on the operational context and estimated serviceability of individual zones. For example, high-severity zone Z11 is prioritized using air transport at timestep 10, reflecting the policy’s tendency to allocate faster transportation modes to critical regions. Second, the rollout includes both successful and failed dispatches. Importantly, failed dispatches occur despite operational feasibility, indicating that the environment incorporates stochastic transition dynamics rather than deterministic routing assumptions. For instance, the dispatch at timestep 26 fails due to capacity exhaustion, demonstrating that the policy must operate under dynamic resource constraints and uncertain infrastructure

availability.

Third, positive timestep rewards are still observed even when some individual dispatches fail. This behavior is expected because the environment reward is computed at the aggregate timestep level, combining humanitarian impact, coverage improvement, equity, operational cost, blockage penalties, and delivery effectiveness. Consequently, localized failures may still yield positive overall utility when other concurrent dispatches successfully reduce unmet demand or improve disaster response coverage. Overall, the rollout demonstrates that the proposed PD-PPO framework learns feasibility-aware, uncertainty-sensitive, and sequentially adaptive dispatch behavior under stochastic disaster conditions.

4.7. Cross-Scenario Evaluation: Thailand Flood Case

To examine cross-scenario transferability, we evaluated the proposed PD-PPO framework on a second disaster environment representing the 2011 Chao Phraya basin floods in Thailand. The training configuration follows Table 1, with the primary modification being a reduced 600-iteration training horizon. The observation and policy structures remain consistent with the Nepal case, while the Thailand environment introduces flood-driven accessibility variation and an additional transport mode (boat), increasing action-space complexity and environment stochasticity.

Training diagnostics in Fig. B.14 show behaviour broadly similar to the Nepal experiments, although the state-critic loss exhibits somewhat higher variance under rapidly changing flood accessibility conditions. Entropy increases during the early training phase and remains comparatively high throughout training, consistent with the expanded multi-modal dispatch space (road-air-boat). Episode rewards show a temporary reduction during the transition to the fully stochastic regime (approximately iterations 60–90) before stabilizing. Policy loss decreases over time while remaining non-zero, suggesting continued policy adaptation under noisy dynamics.

The qualitative policy visualizations in Fig. B.15 indicate that the learned policy adjusts transport-mode usage according to accessibility conditions. Boat and air dispatch actions appear more frequently in highly inundated regions, while road transport continues to be selected in zones with comparatively stable connectivity. Zone-selection heatmaps and action-sequence plots further suggest that the policy reallocates dispatch activity as flood severity evolves across provinces.

Overall, the Thailand experiments indicate that the proposed architecture remains operational under a substantially different disaster scenario involving altered geography, hazard dynamics, and transport structure. While this does not constitute formal out-of-distribution generalization, the results suggest that the recurrent autoregressive PD-PPO framework can adapt to multiple humanitarian logistics settings without requiring changes to the underlying policy architecture.

4.8. Limitations and Future Work

The current framework relies on stylized hazard dynamics, simplified accessibility transitions, and scenario-specific train-

ing, which limits direct deployment in real-world disaster operations. The present implementation also assumes a single-depot setting and does not incorporate live geospatial, traffic, or meteorological data streams.

Although the proposed framework includes reward-sensitivity and ablation analysis, additional large-scale evaluation across broader operational settings and disaster scenarios would further strengthen the robustness assessment. Comparisons with value-based reinforcement learning approaches such as DQN were not included due to the large combinatorial and dynamically masked action space considered in this work.

Future work will extend the framework toward multi-agent and hierarchical coordination settings, integrate real-time GIS and remote-sensing information, investigate transfer learning across disaster types and geographic regions, and incorporate human-in-the-loop feedback mechanisms to improve transparency, adaptability, and operational usability.

5. Conclusion

This work investigated structured actor-critic reinforcement learning for dynamic humanitarian logistics under partial observability, stochastic disruptions, and operational constraints. We introduced an OpenAI Gym-compatible disaster-response simulation framework together with a PD-PPO formulation that integrates autoregressive action factorization, feasibility-aware masking, belief-guided accessibility estimation, and curriculum-controlled stochastic training within a unified sequential decision-making framework. Experimental evaluation across Nepal and Thailand disaster scenarios demonstrated that the proposed approach achieves higher delivered-help performance and improved reward consistency relative to optimization-based and heuristic dispatch baselines under stochastic operating conditions.

While Rolling-LP remains competitive under deterministic policy decoding and paired stochastic environment evaluation, its performance becomes increasingly sensitive to evolving environmental disruptions and computational complexity as problem scale increases. In contrast, PD-PPO maintains stable operational behavior together with substantially lower inference time, highlighting the potential of trained reinforcement-learning policies as adaptive and computationally efficient decision-support mechanisms for time-critical humanitarian operations.

The ablation and sensitivity analyses further suggest that feasibility-aware masking and belief-guided accessibility estimation provide the most consistent practical benefits in large combinatorial dispatch environments characterized by dynamic accessibility and resource uncertainty. More broadly, the results indicate that structured policy-gradient methods can effectively coordinate high-dimensional sequential dispatch decisions without requiring explicit combinatorial optimization during deployment.

Although the proposed framework demonstrates promising performance across multiple disaster settings, several limitations remain. The current environment relies on simulated

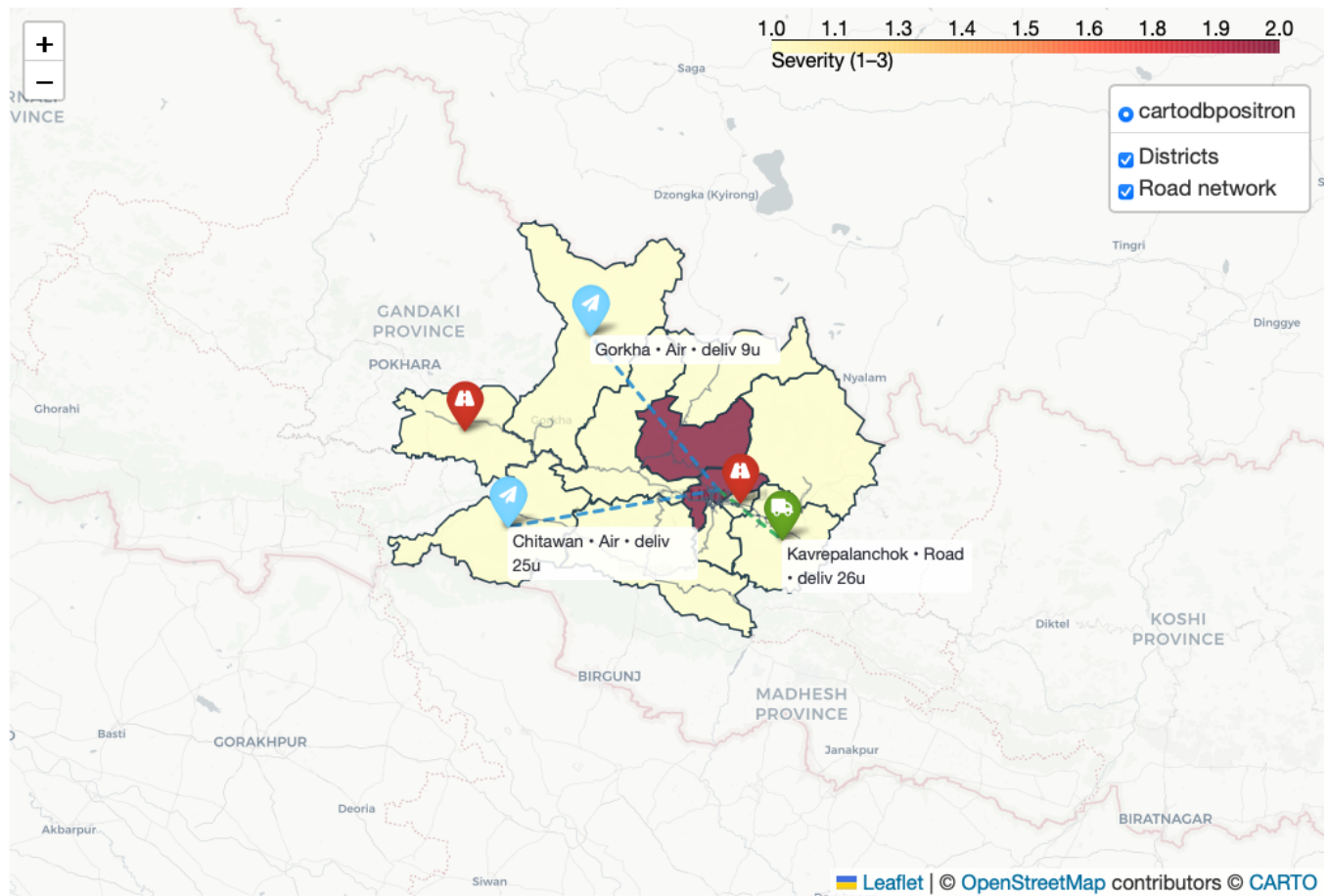
stochastic hazard evolution and simplified infrastructure assumptions, while real-world disaster response often involves richer inter-organizational coordination, communication delays, and evolving human behavioral dynamics. Future work may therefore incorporate real-time operational data streams, richer transportation-network modeling, multi-agent coordination mechanisms, and transfer-learning strategies across heterogeneous disaster environments. Overall, this work provides both a reproducible disaster logistics benchmark environment and empirical evidence that structured actor–critic reinforcement learning can serve as a practical and scalable framework for adaptive humanitarian resource allocation under uncertainty.

Appendix A. Additional Results and Visualizations

Table Appendix A.1: Scalability evaluation across problem sizes. Rewards are normalized; computation times reported in seconds. The slightly lower normalized rewards (typically 0.70–0.80) for small out-of-distribution instances arise because the PD-PPO agent was trained on the full ($Z=12, R=3$) configuration and is therefore evaluated on instances with different spatial structure and resource dimensionality. Despite this distribution shift, the learned policy maintains consistent performance and achieves lower inference time compared with repeated MILP solves, demonstrating strong generalization and computational efficiency.

Instance	Greedy R	LP R	PD-PPO R	Greedy (s)	LP (s)	PPO (s)
Z=5, R=1	0.97	0.90	0.76	0.35	14.10	0.36
Z=9, R=1	0.73	0.91	0.72	0.53	19.10	0.40
Z=12, R=1	0.97	0.88	0.72	0.67	57.04	0.44
Z=5, R=2	0.89	0.83	0.76	0.51	44.63	0.35
Z=9, R=2	0.86	0.96	0.69	0.41	126.65	0.30
Z=12, R=2	1.03	0.94	0.78	1.10	189.91	0.41
Z=5, R=3	1.05	1.06	0.76	0.68	37.76	0.35
Z=9, R=3	0.70	0.79	0.79	1.15	37.35	0.41
Z=12, R=3	0.80	0.88	0.78	1.51	42.89	0.44

Figure A.13: Geographic visualization of unmet demand and resource deliveries during a simulated disaster episode. The system integrates district-level severity, road networks, and mode-specific dispatch decisions (road/air), illustrating the intended real-time decision support use case for humanitarian logistics. Basemap and tiles © OpenStreetMap contributors; rendered using Leaflet and CARTO.



Appendix B. Thailand Training Metric Evaluation

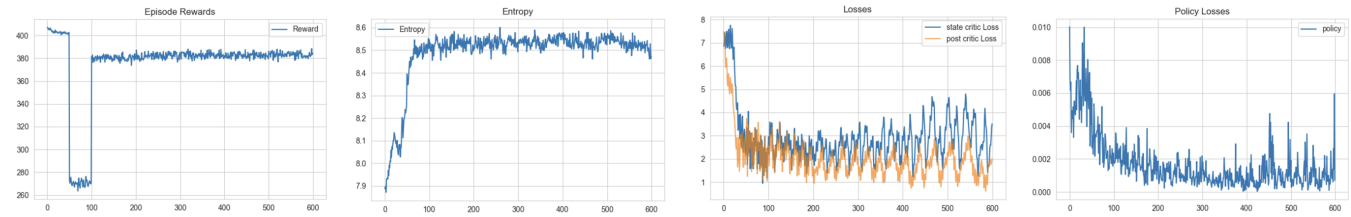


Figure B.14: Training diagnostics for the Thailand flood configuration, including state/post-decision critic losses, episodic returns, entropy, and policy loss over 600 iterations.

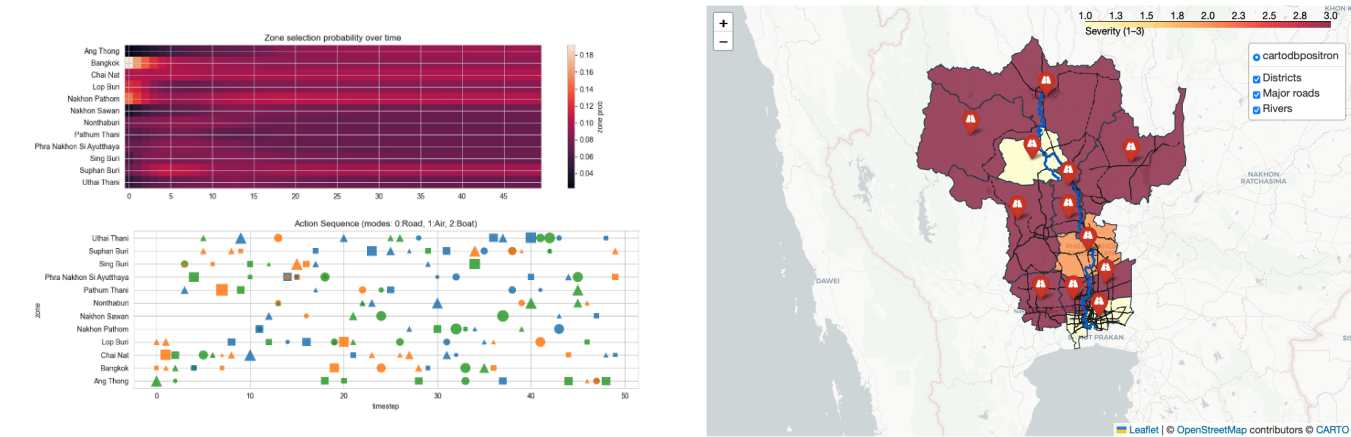


Figure B.15: (a) Policy behavior in the Thailand case: zone-selection probabilities and mode/resource action sequences across representative episodes, and (b) spatial deployment across flood-affected provinces, illustrating multi-modal dispatch under flood conditions.

Appendix C. Baseline Methods

This appendix provides a concise description of the two non-RL baselines used for comparison against the proposed PD-PPO framework. Both baselines operate under the same state representation, dispatch limits, and mode-feasibility rules as the RL agent, ensuring fair evaluation.

C.1. Rolling-LP (Receding-Horizon Mixed-Integer Program)

At each decision step t , we construct a finite set of feasible dispatch candidates

$$i \in \mathcal{I}_t \Leftrightarrow i = (z_i, r_i, m_i, t_i, q_i, u_i),$$

where z_i is the zone, r_i the resource type, m_i the mode (road/air), t_i the number of trips, q_i the quantity bin, and u_i the implied shipment units for that row. Accessibility constraints and per-step feasibility masks are enforced when building $\mathcal{I}_t$ (only open modes and feasible (z, r, t, q) combinations are included).

The Rolling-LP baseline selects a subset of rows via binary decision variables $y_i \in \{0, 1\}$ by maximizing a priority-weighted impact score that mirrors the greedy baseline:

$$\max_{y_i} \sum_{i \in \mathcal{I}_t} w_i y_i, \quad (\text{A.1})$$

where the weight w_i for candidate i is

$$w_i = (\text{gap}_{z_i, r_i, t} s_{z_i, t} \text{pop}_{z_i}) \frac{p_{z_i, m_i, t}}{\eta_{z_i, m_i}} u_i - \lambda_{\text{cost}} c_{m_i} u_i - \lambda_{\text{risk}} (1 - p_{z_i, m_i, t}) u_i. \quad (\text{A.2})$$

Here $\text{gap}_{z, r, t}$ is the normalized demand, $s_{z, t}$ is severity, pop_z a population weight, $p_{z, m, t}$ the estimated probability that mode m is open, $\eta_{z, m}$ an effective travel-time factor, and c_m the per-unit cost (road vs. air). The coefficients λ_{cost} and λ_{risk} (set to $\lambda_{\text{cost}}=0.05$ and $\lambda_{\text{risk}}=0.60$ in our experiments) trade off impact, cost, and routing risk.

The optimization is subject to per-resource and per-demand caps, as well as a limit on the number of distinct dispatch rows executed in a single step:

(depot stock and per-resource dispatch cap)

$$\sum_{i \in \mathcal{I}_t: r_i = r} u_i y_i \leq \min(S_{r, t}, M_{r, t}), \quad \forall r. \quad (\text{A.3})$$

(demand constraint with slack $\varepsilon = 0.05$)

$$\sum_{i \in \mathcal{I}_t: z_i = z, r_i = r} u_i y_i \leq (1 + \varepsilon) D_{z, r, t}, \quad \forall z, r. \quad (\text{A.4})$$

(at most K dispatch rows per step)

$$\sum_{i \in \mathcal{I}_t} y_i \leq K. \quad (\text{A.5})$$

$$y_i \in \{0, 1\}, \quad \forall i \in \mathcal{I}_t. \quad (\text{A.6})$$

Only the first-step solution is executed before the environment transitions and the LP is re-solved at the next decision epoch, yielding a standard receding-horizon optimization baseline that is strong but does not explicitly reason about long-term

exploration or future uncertainty. In our implementation this mixed-integer program is solved using an open-source MILP solver (PuLP with Coin-or branch and cut (CBC), with OR-Tools as a fallback), and the same one-step formulation is used for both deterministic and stochastic environment settings; only the underlying transition dynamics differ.

C.2. Greedy Heuristic (Priority-Weighted Dispatch)

The Greedy baseline evaluates all feasible zone-resource-mode-trip-quantity candidates (z, r, m, t, q, u) generated by the same masking and accessibility rules used in the RL agent and the Rolling-LP formulation. For each candidate $i \in \mathcal{I}_t$, a myopic priority score $\tilde{w}_i$ is computed using the same impact, cost, and risk terms as in the LP objective:

$$\tilde{w}_i = (\text{gap}_{z_i, r_i, t} s_{z_i, t} \text{pop}_{z_i}) \frac{p_{z_i, m_i, t}}{\eta_{z_i, m_i}} u_i - \lambda_{\text{cost}} c_{m_i} u_i - \lambda_{\text{risk}} (1 - p_{z_i, m_i, t}) u_i. \quad (\text{A.7})$$

Candidates are sorted in descending order of $\tilde{w}_i$, and the heuristic selects up to K rows greedily subject to depot stock, per-resource dispatch limits, and per-zone demand feasibility. Letting $U_{r, t}$ denote remaining depot stock and $D_{z, r, t}$ remaining demand, the greedy loop accepts row i whenever

$$u_i \leq U_{r_i, t}, \quad u_i \leq D_{z_i, r_i, t},$$

and updates the residual capacities. No backtracking or global optimization is performed.

This heuristic is extremely fast and reflects operational triage-style decision making, but it is inherently myopic: it does not reason about long-term uncertainty, mode reliability, or interactions between multiple dispatches. It therefore serves as a strong but interpretable non-RL baseline for evaluating the benefits of the PD-PPO policy.

Appendix D. Detailed Statistical Reward Sensitivity & Ablation Results

Table Appendix D.1: Absolute mean performance and 95% confidence intervals across reward-weighting profiles.

Profile	Reward	Help	Cost	Block / Delay
Default	0.731 [0.698, 0.770]	0.465 [0.440, 0.513]	0.096 [0.092, 0.103]	0.007 / 0.013
Efficiency	0.653 [0.608, 0.701]	0.375 [0.323, 0.427]	0.083 [0.080, 0.085]	0.007 / 0.005
Equity	0.697 [0.663, 0.740]	0.427 [0.389, 0.466]	0.090 [0.080, 0.099]	0.005 / 0.017
Humanitarian	0.679 [0.644, 0.717]	0.401 [0.356, 0.444]	0.081 [0.070, 0.101]	0.009 / 0.009
Robustness	0.717 [0.642, 0.819]	0.452 [0.359, 0.548]	0.090 [0.082, 0.098]	0.003 / 0.019

Table Appendix D.2: Paired delta analysis against the default reward configuration. Negative values indicate improvement relative to the default profile for penalty terms.

Profile	Δ Reward	Δ Help	Δ Cost	Δ Delay
Efficiency	-0.078 [-0.162, -0.024]	-0.090 [-0.190, -0.013]	-0.013 [-0.019, -0.009]	-0.007 [-0.014, 0.004]
Equity	-0.034 [-0.083, 0.042]	-0.038 [-0.086, 0.024]	-0.007 [-0.014, -0.002]	0.004 [-0.010, 0.013]
Humanitarian	-0.052 [-0.126, -0.008]	-0.064 [-0.157, 0.004]	-0.016 [-0.032, 0.009]	-0.004 [-0.010, 0.008]
Robustness	-0.014 [-0.083, 0.049]	-0.013 [-0.081, 0.035]	-0.006 [-0.012, 0.004]	0.006 [0.002, 0.011]

Table Appendix D.3: Paired confidence interval analysis for ablation variants relative to the baseline configuration.

Ablation	Metric	Mean Diff	CI Low	CI High
no curriculum	Reward	0.125	-0.040	0.323
	Help	0.129	-0.015	0.304
	Cost	-0.005	-0.008	-0.001
	Block	0.011	0.003	0.018
	Delay	0.024	0.004	0.053
no masks	Reward	0.242	-0.165	0.905
	Help	0.142	-0.047	0.483
	Cost	0.005	-0.008	0.014
	Block	-0.038	-0.093	0.014
	Delay	0.014	-0.003	0.040
no belief	Reward	0.318	0.097	0.543
	Help	0.294	0.198	0.353
	Cost	0.019	-0.012	0.072
	Block	-0.038	-0.117	0.012
	Delay	0.012	-0.012	0.052
no autoregressive	Reward	0.063	-0.060	0.258
	Help	0.051	-0.085	0.266
	Cost	0.016	-0.009	0.030
	Block	-0.008	-0.030	0.004
	Delay	0.011	-0.008	0.045
no dualcritic	Reward	-0.028	-0.178	0.063
	Help	0.009	-0.114	0.104
	Cost	-0.001	-0.014	0.007
	Block	0.003	-0.006	0.012
	Delay	0.019	0.000	0.052

Appendix E. Statistical Significance Table of Baseline Comparison

Table Appendix E.4: Statistical comparison of PD-PPO against deterministic baselines across 10 stochastic evaluation seeds and 500 paired rollout episodes. Bootstrap confidence intervals (CI) report the mean performance difference (PD-PPO minus baseline). Statistical significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Metric	Comparison	Best	t -test	Wilcoxon	Cohen's d	Bootstrap CI	Interpretation
Reward	PD-PPO vs LP	PD-PPO	< 0.001***	< 0.001***	0.278	[0.040, 0.076]	Higher reward
Reward	PD-PPO vs Greedy	PD-PPO	< 0.001***	< 0.001***	0.261	[0.031, 0.063]	Higher reward
Help	PD-PPO vs LP	PD-PPO	< 0.001***	< 0.001***	0.549	[0.097, 0.133]	Substantially higher aid
Help	PD-PPO vs Greedy	PD-PPO	< 0.001***	< 0.001***	0.547	[0.083, 0.115]	Substantially higher aid
Cost	PD-PPO vs LP	LP	0.199	0.034*	0.058	[-0.001, 0.006]	Comparable cost
Cost	PD-PPO vs Greedy	Greedy	0.505	0.013*	0.030	[-0.002, 0.005]	Comparable cost
Block	PD-PPO vs LP	LP	< 0.001***	< 0.001***	0.308	[0.003, 0.006]	More aggressive dispatch
Block	PD-PPO vs Greedy	Greedy	< 0.001***	< 0.001***	0.447	[0.005, 0.007]	More aggressive dispatch
Delay	PD-PPO vs LP	LP	0.330	0.045*	0.044	[-0.002, 0.006]	Comparable delay
Delay	PD-PPO vs Greedy	PD-PPO	0.177	0.413	-0.061	[-0.006, 0.001]	Comparable delay

Appendix F. Training Stability Diagnostics

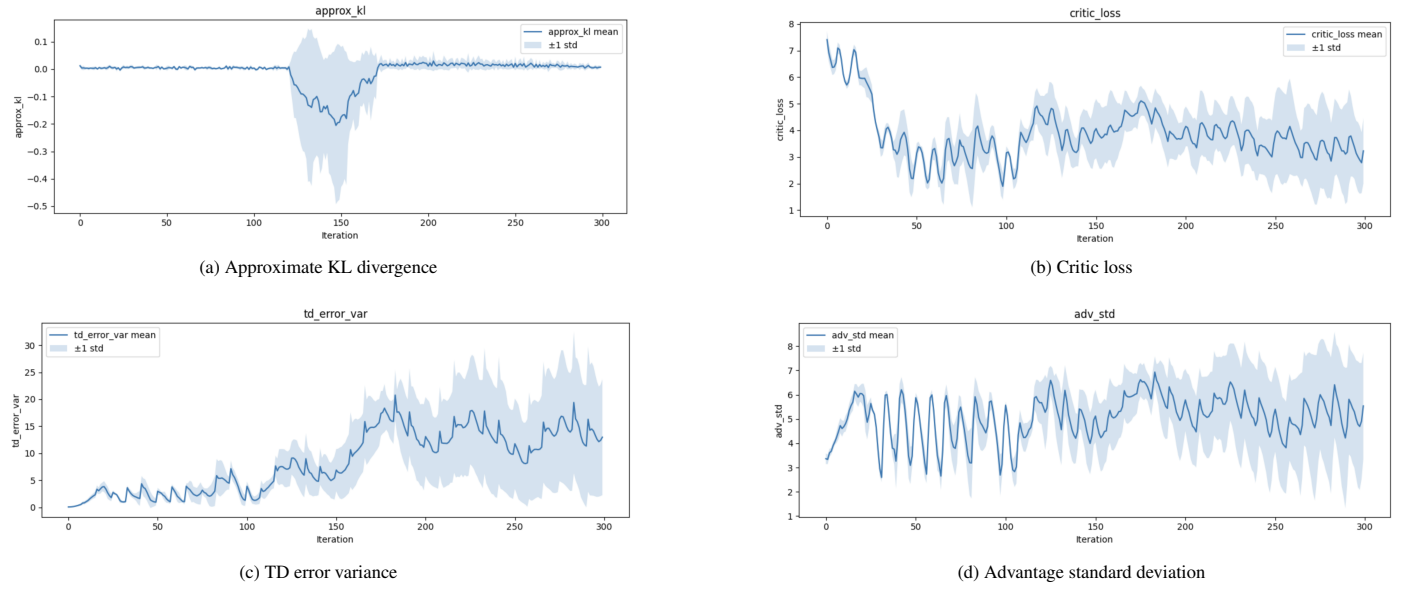


Figure F.16: Training diagnostics of the proposed PD-PPO model across five random seeds. Solid lines indicate the mean value and shaded regions represent one standard deviation. The plots illustrate optimization behavior during curriculum learning, including policy update magnitude (KL divergence), critic estimation quality, TD-error variability, and advantage dispersion.

Table Appendix F.5: Final performance and optimization statistics across five seeds (mean $\pm$ standard deviation).

Metric	Mean	Std
Final Iteration Reward	187.744	0.835
Final Reward	0.673	0.018
R_{help}	0.432	0.015
R_{cov}	0.284	0.001
R_{deq}	0.013	0.005
P_{cost}	0.078	0.002
P_{failU}	0.0004	0.0002
P_{block}	0.0002	0.0002
P_{noops}	0.0000	0.0000
Time Delay Penalty	0.0026	0.0006
Approximate KL	0.0067	0.0056
TD Error Variance	13.395	11.359
Advantage Std	5.183	2.342
Critic Loss	3.271	1.328

Appendix G. Stochastic Transition Components

These components parameterize the primary stochastic transition mechanisms of the environment.

Initial Severity Sampling: At the beginning of each episode, zone severity levels are initialized using a hazard-dependent categorical distribution:

$$h_{z,0} \sim \text{Categorical}(\pi_z), \quad (\text{G.1})$$

where the probability vector π_z depends on the hazard exposure level of zone z . High-hazard zones are assigned larger probability mass toward severe states.

In deterministic mode, severity is assigned using threshold rules:

$$h_{z,0} = \begin{cases} 3, & e_z > 0.66, \\ 2, & 0.33 < e_z \leq 0.66, \\ 1, & \text{otherwise}, \end{cases}$$

where e_z denotes normalized hazard exposure.

Condition-State Transitions: Zone condition dynamics are governed by terrain-dependent categorical transition matrices:

$$c_{z,t+1} \sim \text{Categorical}(P_c^{(\tau_z)}(c_{z,t}, \cdot)), \quad (\text{G.2})$$

where $\tau_z \in \{\text{plain, valley, hilly}\}$ denotes terrain type.

The condition states are:

$$c_{z,t} \in \{\text{Normal, HeavyRain, Landslide, Both}\}.$$

In deterministic mode, the next condition state is selected using:

$$c_{z,t+1} = \arg \max_j P_c^{(\tau_z)}(c_{z,t}, j).$$

Cooldown Sampling: After a condition transition occurs, each zone samples a cooldown interval before another transition becomes possible:

$$\delta_{z,t} \sim \text{DiscreteUniform}(l_{z,t}, u_{z,t}), \quad (\text{G.3})$$

where the interval bounds $(l_{z,t}, u_{z,t})$ depend on terrain category, hazard persistence, and annealing schedules.

In deterministic mode, the cooldown is set to the interval midpoint:

$$\delta_{z,t} = \left\lfloor \frac{l_{z,t} + u_{z,t}}{2} \right\rfloor.$$

Air-Grounding Uncertainty: Under heavy-rain or combined-hazard conditions, air transport may become temporarily unavailable. Air-grounding events are modeled as Bernoulli random variables:

$$G_{z,t} \sim \text{Bernoulli}(p_{z,t}^{\text{air}}), \quad (\text{G.4})$$

where $G_{z,t} = 1$ denotes a grounded air route.

The grounding probability depends on terrain and hazard exposure:

$$p_{z,t}^{\text{air}} = g(c_{z,t}, \tau_z, e_z), \quad (\text{G.5})$$

where $g(\cdot)$ increases under heavy rain, difficult terrain, and higher hazard exposure.

Restock and Aftershock Timing: The midpoint international-aid restock trigger and aftershock trigger are sampled from discrete uniform distributions:

$$t_{\text{restock}} \sim \text{DiscreteUniform}\left(\frac{T}{2} - 5, \frac{T}{2} + 5\right), \quad (\text{G.6})$$

$$t_{\text{aftershock}} \sim \text{DiscreteUniform}(3, 9), \quad (\text{G.7})$$

where T denotes the episode horizon.

In deterministic mode, these triggers are fixed at:

$$t_{\text{restock}} = \frac{T}{2}, \quad t_{\text{aftershock}} = 6.$$

Aftershock Magnitude Sampling: When aftershock events occur, demand surges are scaled using multiplicative uniform noise:

$$M_{z,t} \sim \text{Uniform}(1.2, 1.4). \quad (\text{G.8})$$

The sampled multiplier increases unmet demand and temporarily elevates severity levels in affected zones. Severity evolution additionally incorporates bounded stochastic perturbations driven by unmet demand growth, delayed service, and exogenous disruption events.

Priority Noise: Initial priority estimation includes additive Gaussian perturbations:

$$\epsilon_z \sim \mathcal{N}(0, 0.05^2), \quad (\text{G.9})$$

which introduces mild stochastic variability in zone-priority rankings.

In deterministic mode:

$$\epsilon_z = 0.$$

Initial Demand Jitter: Initial resource demand is perturbed using multiplicative uniform noise:

$$J_{z,r} \sim \text{Uniform}(0.85, 1.15), \quad (\text{G.10})$$

such that initialized demand becomes:

$$D_{z,r,0} = \bar{D}_{z,r,0} \cdot J_{z,r}, \quad (\text{G.11})$$

where $\bar{D}_{z,r,0}$ denotes the baseline demand derived from severity, remoteness, hazard exposure, and socio-economic vulnerability factors.

Deterministic Transition Approximation. In deterministic mode, all stochastic sampling operators are replaced by their deterministic counterparts:

- categorical transitions use $\arg \max$ selection,
- Bernoulli variables use thresholded deterministic outcomes,
- discrete-uniform cooldowns use interval midpoints,
- Gaussian perturbations are removed,
- stochastic event triggers are replaced by fixed schedules.

This preserves the structural environment dynamics while removing transition variance, enabling controlled reproducibility experiments.

Appendix H. Implementation and Architecture Details

Table Appendix H.6: Neural architecture and implementation details of the proposed PD-PPO framework.

Component	Configuration
Observation representation	Flattened environment observation vector of dimension d_{obs} including zone demand, severity, accessibility, inventory, priority, and transport-state features
Encoder network	Two-layer MLP: $\text{Linear}(d_{\text{obs}}, 256) \rightarrow \text{ReLU} \rightarrow \text{Linear}(256, 256) \rightarrow \text{ReLU}$
Recurrent memory module	3-layer LSTM with hidden size 256, dropout 0.15, batch-first recurrent processing
LSTM hidden state	$h_t^{\text{LSTM}} \in \mathbb{R}^{256}$ per layer; hidden and cell states propagated across timesteps
Feature normalization	LayerNorm(256) applied after recurrent encoding
Autoregressive policy structure	Sequential conditional factorization: Zone $\rightarrow$ Mode $\rightarrow$ Resource $\rightarrow$ Trips $\rightarrow$ Quantity
Action heads	Zone head: $\text{Linear}(256, Z)$ Mode head: $\text{Linear}(256, M)$ Resource head: $\text{Linear}(256, R)$ Trips head: $\text{Linear}(256, T)$ Quantity head: $\text{Linear}(256, 128) \rightarrow \text{GELU} \rightarrow \text{Linear}(128, Q)$
Conditional action embeddings	Learned conditioning projections: $\text{Linear}(Z, 256)$, $\text{Linear}(M, 256)$, $\text{Linear}(R, 256)$, $\text{Linear}(T, 256)$
Feasibility masking	Dynamic masking applied at each autoregressive decision stage using environment feasibility constraints and accessibility conditions
Belief module	Accessibility belief head: $\text{LayerNorm}(256) \rightarrow \text{Linear}(256, 256) \rightarrow \text{SiLU} \rightarrow \text{Dropout}(0.1) \rightarrow \text{Linear}(256, 256)$ with residual connection
Belief output	$\text{Linear}(256, 2Z)$ producing road/air accessibility logits for each zone
State critic	$\text{Linear}(256, 1)$ estimating state value $V_\phi(o_t)$
Post-decision critic	$\text{Linear}(256, 128) \rightarrow \text{ReLU} \rightarrow \text{Linear}(128, 1)$ estimating $V_\phi^{PD}(o_t, a_t)$
Initialization	Orthogonal initialization for recurrent hidden weights and critic layers; Xavier initialization for feed-forward layers; LSTM forget-gate bias initialized to 1.0
Activation functions	ReLU (encoder and critic), GELU (quantity head), SiLU (belief module), Tanh (conditioning projections)
Optimization stabilizers	Gradient clipping (max norm 0.4), MAD-based advantage normalization, KL-penalized PPO updates, entropy regularization, curriculum-controlled stochasticity

Note: A three-layer recurrent architecture was selected empirically to balance temporal representation capacity and optimization stability under long-horizon stochastic dynamics.

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